The role of thermodynamic forcing in direct numerical simulations of turbulence-cloud microphysics interactions

Abdullah Al Muti Sharfuddin,[1] Foluso Ladeinde,[1,a)] Yangang Liu,[2, b)] Satoshi Endo,[2] Fan Yang,[2] Mohammad Atif,[3] Meifeng Lin,[3] and Wenhai Li[4]

[1]*Department of Mechanical Engineering, 113 Light Engineering Building, Stony Brook University, 100 Nicolls Road, Stony Brook, New York, 11794-2300, USA*

[2]*Environmental Science and Technologies Department, Brookhaven National Laboratory, Rutherford Dr, Upton, New York, 11973-5000, USA*

[3]*Computing and Data Sciences Directorate, Brookhaven National Laboratory, 2 Center St, Upton, New York, 11973-5000, USA*

[4]*Department of Mechanical Engineering Technology, Farmingdale State College, 2350 Broadhollow Road, Farmingdale, New York, 11735-1021, USA*

Particle-based direct numerical simulation (DNS) is a promising approach to study microscale turbulence-cloud-aerosol interactions. Such DNS models require external forcing to sustain turbulent fluctuations in the momentum and thermodynamic fields. Although momentum forcing is common in particle-based DNS studies of atmospheric clouds, thermodynamic forcing has not gained enough attention. This study implements external forcing in the thermodynamic fields of temperature and vapor mixing ratio in addition to momentum forcing. The forcing terms are constructed in the Fourier space using a band of low wavenumbers. The turbulent flow of air is assumed to be homogeneous and isotropic. Aerosol particles and cloud droplets are tracked in Lagrangian fashion, with processes of condensation, evaporation, activation, and deactivation considered together. The impacts of thermodynamic forcing on entrainment of ambient air into the cloud and subsequent turbulent mixing are examined. It is found that thermodynamic forcing sustains fluctuations in the thermodynamic fields and reduces the mean temperature, water vapor mixing ratio, and supersaturation. Although thermodynamic forcing is applied at large scales only, it influences the energy at all scales by interacting with latent heat change. The probability distributions of forced thermodynamic fields are negatively skewed with kurtosis values larger than three. Further analysis of microphysical properties shows that thermodynamic forcing enhances evaporation and deactivation with broadening of particle size distributions. The activation and deactivation regimes are fluctuation-dominated if thermodynamic forcing is present as opposed to being mean-dominated only when thermodynamic forcing is not applied.

[a), b)] Author to whom correspondence should be addressed. Electronic mail: a) Foluso.Ladeinde@Stonybrook.edu, b) lyg@bnl.gov

NOMENCLATURE

c_p, C_d	= specific heat of air [J/(kg.K)], time rate of exchange of mass [1/s]
δ_{k,k_f}, η	= delta function [-], Kolmogorov length scale [m]
$\epsilon_{T,in}$, $\epsilon_{q_v,in}$	= dissipation rates in the initial temperature [K^2/s], and vapor [$(g/kg)^2/s$] fields
ϵ	= turbulent kinetic energy dissipation rate [m^2/s^3]
$\boldsymbol{F}_b$, $\boldsymbol{F}_c$	= buoyancy force [m/s^2], external force [m/s^2] in the momentum field per unit mass
F_T, F_{q_v}	= external forcing in the temperature [K/s], water vapor [(g/kg)/s] fields
$\boldsymbol{g}$, $\boldsymbol{k}_f$	= acceleration vector due to gravity [m/s^2], forcing wavenumber vector [1/m]
k_{min}, k_{max}	= minimum wavenumber [1/m], maximum wavenumber [1/m]
κ, k_T	= hygroscopicity of solute [-], thermal conductivity in air [W/(m.K)]
l_0, l	= length scale characterizing large eddies [m], integral length scale [m]
L_h, λ, M_l	= specific latent heat [J/kg], Taylor microscale [m], molar mass of water [kg/mol]
m_a, μ_T, μ_v	= mass of air [kg], molecular diffusivities [m^2/s] of temperature, water vapor in air
ν, p	= kinematic viscosity of air [m^2/s], instantaneous fluid pressure [N/m^2]
q_v, $q_{v,s}$	= instantaneous water vapor mixing ratio [g/kg], saturation vapor mixing ratio [g/kg]
r_d, r	= dry aerosol radius [μm], wet radius of aerosol particle or cloud droplet [μm]
r_c, R_v	= critical radius [μm], specific gas constant for water vapor [J/(kg.K)]
$R_{q'_v(F_{q_v}-C_d)'}$	= correlation coefficient between fluctuations in water vapor and the source term [-]
$R_{S'_e w'}$	= correlation coefficient between fluctuations in supersaturation and z-velocity [-]
ρ_a, ρ_l, S_e	= density of air [kg/m^3], density of water [kg/m^3], environmental supersaturation [-]
S_k, S_c	= particle equilibrium supersaturation [-], critical equilibrium supersaturation [-]
S_t, σ_l	= Stokes number [-], surface tension of water [N/m]
T, τ_p	= instantaneous temperature [K], particle response time [s]
$\boldsymbol{u}$	= instantaneous fluid velocity vector [m/s]
$\boldsymbol{V}$, $\boldsymbol{X}$	= particle velocity vector [m/s], particle position vector [m]
PSD	= particle size distribution [$\mu m^{-1} cm^{-3}$]
PDF, TKE	= probability density function [1/unit], turbulent kinetic energy [m^2/s^2]

I. INTRODUCTION

Numerical modeling has become a powerful tool to investigate turbulence-cloud-aerosol interactions in the Earth's atmosphere as experiments and measurements are difficult and expensive to conduct. The most accurate approach is direct numerical simulation (DNS) coupled with Lagrangian particle tracking (Liu et al.[1]). This method resolves the flow of turbulent air upto the smallest scales without any kind of parameterizations of small turbulent eddies. It also tracks cloud droplets and aerosol particles individually. Two thermodynamic (scalar) fields, namely temperature and water vapor mixing ratio, are transported in the Eulerian framework. Flow turbulence is assumed to be homogeneous and isotropic, but it decays in the absence of a turbulence production mechanism. Many DNS studies (Vaillancourt,[2] Kumar et al.,[3] Li[4]) and our previous DNS studies (Gao et al.,[5] Sharfuddin et al.[6]) have applied external forcing in the momentum field to maintain turbulence. However, thermodynamic fluctuations decayed in those studies. In the atmosphere, thermodynamic fluctuations are sustained by mechanisms such as the entrainment of ambient air into the cloud and solar radiation absorbed by the cloud.[7,8,9] Numerical representation of these events requires the addition of external forcing in the thermodynamic fields.

The idea of forcing is to inject energy to a band of low wavenumber in the field of interest from where they are transferred to the high wavenumber region of the turbulence spectrum. Eswaran and Pope[10] showed that Fourier forcing in the momentum field results in a statistically stationary velocity field. Bogucki et al.,[11] Watanabe and Gotoh,[12] and Chen and Cao[13] applied Fourier forcing in a passive scalar (thermodynamic) field. They suggest that the scaling behavior of a passive scalar differs from that of the velocity field because scalar fluctuations are highly intermittent at small scales and become decorrelated more rapidly than velocity fluctuations do. In the context of atmospheric clouds, the DNS model of Paoli and Shariff[14] implemented Fourier forcing in the momentum, temperature, and water vapor fields. The authors reported that the temperature and water vapor fields reach statistical stationarity due to the forcing. But their model's setup did not account for latent heat release and turbulent entrainment-mixing processes. The DNS model of Chen et al.[15] forced the thermodynamic fields, also in a low wavenumber band, to simulate the growth of mixed-phase clouds. Saito et al.[16] applied Fourier forcing in the water vapor field but kept the temperature field unforced.

Thermodynamic forcing influences cloud microphysics in numerical models. Paoli and Shariff[14] suggest that the temperature and vapor fluctuations sustained by forcing broaden the size distributions of cloud droplets. Saito et al.[16]

found that droplet size distributions are narrower in their DNS model relative to those observed in the experiments if thermodynamic forcing is absent, but that there is better agreement with forcing. The DNS model of Thomas et al.,[17] in which the adiabatic lapse rate term acted as temperature forcing and the water vapor field was unforced, showed that supersaturation fluctuations depend on temperature forcing and affect the condensation of cloud droplets. Using a similar DNS model, Saito and Gotoh[18] reported that the microphysical fluctuations are strongly impacted by large-scale turbulence. MacMillan et al.[19] conducted a DNS investigation of a laboratory cloud chamber, with the sidewalls acting as a source of forcing in the water vapor field. They found that the probability distributions of supersaturation are non-Gaussian and the size distributions of cloud droplets broaden or narrow depending on the details of simulation parameters.

Despite the abovementioned studies, knowledge gaps still exist regarding the role of thermodynamic forcing in microscale cloud models. Specifically, the interaction between latent heat release or absorption and thermodynamic forcing requires better understanding. Also, microphysical events like condensation, aerosol activation, evaporation and cloud droplet deactivation need to be understood under sustained thermodynamic fluctuations. In this paper, we extend our previous DNS study[6] to include external forcing in the temperature and vapor mixing ratio fields. The forcing is carried out at the low wavenumbers using the approach of Fourier forcing.[10] Note that thermodynamic forcing is added in addition to the forcing of the base momentum field. The inter-conversion of aerosol particles and cloud droplets is modeled using the kappa-Kohler theory.[20] Simulations are conducted in three spatial dimensions and time for two sets of initial conditions. The forced and unforced thermodynamic fields are compared in terms of the mean, variance, energy spectrum, correlation, and probability distribution. The effects of varying the magnitudes of the critical thermodynamic and momentum forcing metrics are investigated. The microscale processes of condensation and evaporation are studied along with different regimes of activation and deactivation. The impact of external forcing on the broadening and narrowing of particle size distributions is also examined for underlying physical mechanisms.

II. GOVERNING EQUATIONS

The equations solved in our model are described in this section. They include the Navier-Stokes equations for the flow field, the conservation equations for the thermodynamic variables, the models for the Lagrangian particle tracking, microphysical equations, and the kappa-Kohler theory which models the curvature and solute effects.

A. The velocity field

The turbulent flow of ambient air is assumed to be governed by the incompressible Navier-Stokes equations with the Boussinesq approximation:

$$\frac{\partial \boldsymbol{u}}{\partial t} + (\boldsymbol{u}.\nabla)\boldsymbol{u} = -\frac{1}{\rho_a}\nabla p + \nu\nabla^2\boldsymbol{u} + \boldsymbol{F}_b + \boldsymbol{F}_c\,, \tag{1}$$

$$\nabla.\boldsymbol{u} = 0\,. \tag{2}$$

Here, $\boldsymbol{u} \equiv (u, v, w)$ is the instantaneous fluid (air) velocity vector, p is the instantaneous fluid static pressure, and $\boldsymbol{F}_b$ is the buoyancy force which is implemented in the form:[21]

$$\boldsymbol{F}_b = -\boldsymbol{g}\left[\frac{T - \langle T\rangle}{\langle T\rangle} + 0.608(q_v - \langle q_v\rangle) - q_c\right], \tag{3}$$

where $\boldsymbol{g}$ is the acceleration vector due to normal gravity, T is the instantaneous temperature, q_v is the instantaneous water vapor mixing ratio and q_c is the liquid water mixing ratio. The $\langle T\rangle$ and $\langle q_v\rangle$ are the domain averages of T and q_v, respectively. Our model's domain size (0.512 m) is much smaller compared to an actual cloud and thus $\boldsymbol{F}_b$ is found to be relatively insignificant and unable to maintain flow turbulence via buoyancy production. As a result, the momentum field is forced by adding the $\boldsymbol{F}_c$ in Eq. (1). The $\boldsymbol{F}_c$ is formulated in the Fourier space as:[22]

$$\hat{\boldsymbol{f}}_c(\boldsymbol{k}, t) = \epsilon_{in}\frac{\hat{\boldsymbol{u}}(\boldsymbol{k}, t)}{\sum_{k_f \epsilon_k}\left|\hat{\boldsymbol{u}}(\boldsymbol{k}_f, t)\right|^2}\delta_{k,k_f}\,. \tag{4}$$

where ϵ_{in} is an input turbulent kinetic energy (TKE) dissipation rate, $\boldsymbol{k}$ is the wavenumber vector, $\hat{\boldsymbol{u}}(\boldsymbol{k}, t)$ is the velocity field in the Fourier space, $\boldsymbol{k}_f$ is the forced wavenumber vector, and δ_{k,k_f} is a delta function which is unity inside the forced wavenumber shell but zero otherwise. The $\hat{\boldsymbol{f}}_c$ is inverse-transformed into the physical space as $\boldsymbol{F}_c$ in Eq. (1). Note that the "back and forth" with the Fourier and physical space is necessary because our numerical procedure takes place in the physical space, while the subset of the spectrum that is forced can only be directly obtained in Fourier space.

B. Conservation of the thermodynamic fields

The conservation equations for temperature (T) and water vapor mixing ratio (q_v) are solved in the following form:

$$\frac{\partial T}{\partial t} + (\boldsymbol{u}.\nabla)T = \frac{L_h}{c_p}C_d + \mu_T\nabla^2 T + F_T\,, \tag{5}$$

$$\frac{\partial q_v}{\partial t} + (\boldsymbol{u}.\nabla) q_v = -C_d + \mu_v \nabla^2 q_v + F_{q_v} , \tag{6}$$

where C_d quantifies the time rate of exchange between liquid water and water vapor and is described by Eq. (17). The phase change terms in Eqs. (5) and (6) are $L_h C_d / c_p$ and $-C_d$, respectively, where L_h is the specific latent heat and c_p is the specific heat of air. The molecular diffusivities of temperature and water vapor are μ_T and μ_v, respectively, while the F_T and F_{q_v} are the external forcing terms in these equations. Analogous with momentum forcing, the forcing of a generic fluctuating thermodynamic field θ, F_θ, can be written in the Fourier space as:

$$\hat{f}_\theta(\boldsymbol{k}, t) = \epsilon_{\theta,in} \frac{\hat{\theta}(\boldsymbol{k}, t)}{\sum_{k_f \epsilon_k} \left|\hat{\theta}(\boldsymbol{k}_f, t)\right|^2} \delta_{k,k_f} , \tag{7}$$

where $\hat{\theta}(\boldsymbol{k}, t)$ is the fluctuating thermodynamic field in the Fourier space and $\epsilon_{\theta,in}$ is an input dissipation rate for θ. The $\hat{f}_\theta(\boldsymbol{k}, t)$ is converted to the physical space as $F_\theta(\boldsymbol{x}, t)$ after selecting the wavenumbers that are forced. The domain-average of F_θ is zero. The external forcing terms in the momentum and thermodynamic fields are randomly generated and are independent of each other. Note that the impact of adiabatic lapse rate on temperature change is also examined and is found to be negligible (see Appendix A for discussion).

C. Dynamics of cloud and aerosol particles

The motion of each aerosol particle or cloud droplet is modeled in the Lagrangian fashion:[3]

$$\frac{d\boldsymbol{X}_i(t)}{dt} = \boldsymbol{V}_i(t) , \tag{8}$$

$$\frac{d\boldsymbol{V}_i(t)}{dt} = \frac{1}{(\tau_p)_i} [\boldsymbol{u}_i(\boldsymbol{X}, t) - \boldsymbol{V}_i(t)] + \boldsymbol{g} , \tag{9}$$

where the subscript 'i' denotes the i-th aerosol particle or cloud droplet, $\boldsymbol{V}$ is its velocity vector, and $\boldsymbol{X}$ is the position vector. The $\boldsymbol{u}_i$ is the instantaneous fluid velocity vector at the position of particle i and is obtained through a trilinear interpolation of the Eulerian field. The particle response time τ_p is given by[3]

$$(\tau_p)_i = \frac{2\rho_l r_i^2}{9\rho_a \nu} . \tag{10}$$

where ρ_l is the density of liquid water, ρ_a is the air density, and ν is the kinematic viscosity of air. Equation (9) is valid when the particle Stokes number (S_t) is significantly less than unity. The time evolution of particle radius r is described by[23]

$$\frac{dr_i(t)}{dt} = \frac{G_i}{r_i(t)}\big(S_e(\boldsymbol{X},t) - S_k(\boldsymbol{X},t)\big)_i . \tag{11}$$

where S_e is the environmental supersaturation and S_k is the particle equilibrium supersaturation that accounts for the combined curvature and solute effects. The kappa-Kohler theory[20] expresses the S_k as:

$$S_k = \frac{A}{r} - \frac{\kappa r_d^3}{r^3 - r_d^3(1-\kappa)} , \qquad A = \frac{2\sigma_l M_l}{RT\rho_l} , \tag{12}$$

where r_d is the dry aerosol radius, κ is the hygroscopicity of solute (dry aerosol), R is the universal gas constant, σ_l is the surface tension of water, and M_l is molar mass of water. If the atmosphere is supersaturated, aerosol particles grow beyond a threshold, known as the critical radius, and activate into cloud droplets. Note that a particle experiences non-equilibrium, diffusion-limited growth once it passes the critical barrier.[24] In a subsaturated environment, cloud droplets shrink below the critical radius and deactivate into aerosol particles. Using the kappa-Kohler theory, the critical radius r_c and critical equilibrium supersaturation S_c are derived as:[25]

$$r_c = \sqrt{\frac{3\kappa r_d^3}{A}} , \qquad S_c = \frac{2}{\sqrt{\kappa}}\left(\frac{A}{3r_d}\right)^{\frac{3}{2}} . \tag{13}$$

A particle can be classified as aerosol or cloud by comparing its wet radius (r) with r_c. An aerosol particle corresponds to $r_d < r \leq r_c$ while a cloud droplet corresponds to $r > r_c$. The variable S_e depends on the variables q_v and T and is expressed as:

$$S_e = \frac{q_v}{q_{v,s}} - 1 , \quad q_{v,s} = 621.97\frac{p_{sat}}{p - p_{sat}} , \tag{14}$$

$$p_{sat} = 611.2 \times \exp\left(\frac{17.67(T - 273.15)}{T - 29.65}\right) . \tag{15}$$

$q_{v,s}$ is the saturation vapor mixing ratio[26] and p_{sat} is the saturation vapor pressure.[27] The growth parameter G in Eq. (11) is found as:[23]

$$G = \frac{1}{\frac{L_h\rho_l}{k_T T}\left(\frac{L_h}{R_v T} - 1\right)(1 + S_k) + \frac{\rho_l R_v T}{\mu_v p_{sat}}} , \tag{16}$$

where k_T is the thermal conductivity in air and R_v is the universal gas constant divided by the molecular weight of water. The C_d is determined as:[6]

$$C_d(\boldsymbol{x},t) = \frac{4\pi\rho_l}{\rho_a a^3}\sum_{i=1}^{n} G_i r_i(t)\big(S_e(\boldsymbol{X},t) - S_k(\boldsymbol{X},t)\big)_i , \tag{17}$$

where n is the number of particles in a grid cell that has a volume of a^3. In the present study, the value of n is 0.95 and that of a is 0.512 m.

III. NUMERICAL DETAILS

We run simulations for two scenarios: Scenario I and Scenario II.[6] Since the domain size of our DNS model is much smaller than that of a natural cloud, it will be difficult to observe significant condensation and evaporation at the same time within our model volume. As a result, we designed two separate scenarios that correspond to different initial conditions. In the first scenario, the cloud volume mixes with saturated environmental air, and activation and condensation prevail. In the second, the dry (subsaturated) air is entrained into the cloud volume and evaporation and deactivation are dominant.

A. Momentum, thermodynamic, and particle fields

To examine the influences of turbulence intensity, we vary the range of forced wavenumbers to generate two levels of flow turbulence: 'low' and 'high,' which we denote as 'L' and 'H,' respectively.[6] The forcing wavenumber bands are: $1 \le k_f/k_{min} \le 4.12$ and $1 \le k_f/k_{min} \le 16.18$, for the two levels of turbulence. The minimum wavenumber $k_{min} = 2\pi/L = 2\pi/0.512 \sim 12.27$ and the maximum wavenumber $k_{max} \sim 128k_{min}$. The initial TKE spectrum is specified as[28]

$$E(k) = \frac{16}{\sqrt{\pi/2}}\frac{u_0^2 k^4}{k_0^5}\exp\left(-\frac{2k^2}{k_0^2}\right). \tag{18}$$

where $k_0 = 4.12k_{min}$ and $u_0 = 0.16\, m/s$ is the root-mean-square (rms) velocity from the generated fluctuating velocity field. The form k^4 for low wavenumbers corresponds to the incompressible limit while the exponential function allows a rapid roll-off of the energy at high wavenumbers. Table I lists the statistics of the (statistically) stationary turbulence field. The data includes the TKE dissipation rate, ϵ, the Taylor microscale, λ, the Kolmogorov length scale η, the length scale characterizing large eddies, l_0, the forcing Reynolds number, Re_f, the Taylor-scale Reynolds number, Re_λ, and the Reynolds number based on l_0, Re_{l_0}. The definition of abovementioned quantities can be found in Tennekes and Lumley.[29] The values in Table I match the parameters of ambient turbulence as measured by Siebert et al.[30]

Table I. Statistics of the turbulent velocity field at 4 s

Turbulence level	ϵ (m^2/s^3)	λ (m)	η (m)	l_0 (m)	Re_f	Re_λ	Re_{l_0}
High	0.00723	0.0169	0.00082	0.184	455.41	156.11	1704
Low	0.00063	0.0261	0.00153	0.227	201.90	138.74	1207

Our model is that of a 3D cube inside which flow turbulence, cloud droplets, and aerosol particles interact. The boundary conditions are triply periodic, following the standard procedure of DNS. The domain volume is $0.512^3\ m^3$. At the start of simulation, the domain is divided equally into cloudy and clear-air regions. A slab-like configuration[4] is assumed for the initial field (Fig. 1) to enable a sharp transition in the water vapor mixing ratio and temperature at the cloud/clear-air interface. The initial thermodynamic fields change abruptly across the interface in our model unlike those in Paoli and Shariff[14] and Chen et al.[15] Such an inhomogeneous distribution of the initial thermodynamic fields allows us to realistically simulate turbulent mixing between the cloud and environment.[31] However, this configuration precludes the use of the pseudospectral method, which otherwise gives the Gibbs phenomenon at the interfaces,[2] which is unphysical. This partially explains the use of the finite difference method[19] in the current study.

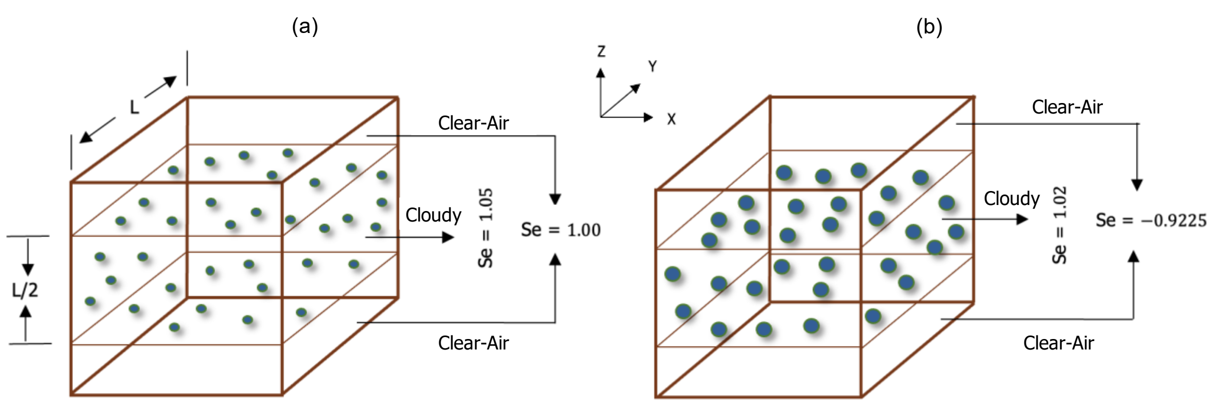


FIG. 1 Initial configuration in (a) Scenario I and (b) Scenario II. The middle half represents the cloudy region while the top and bottom one-fourths constitute the clear-air region. A sharp interface separates the two regions. The cloudy region is initially 5% supersaturated ($S_e = 1.05$) in Scenario I and 2% ($S_e = 1.02$) in Scenario II. The clear-air region is just saturated in the first ($S_e = 1.00$) and 92.25% subsaturated ($S_e = -0.9225$) in the second. The spheres represent dry aerosols in Scenario I and cloud droplets in Scenario II.

In Scenario I, at the initial time, the cloudy region is supersaturated ($S_e = 1.05$), while the clear-air region is just saturated ($S_e = 1.00$). To maintain the distinction between cloud and clear air, the initial vapor mixing ratio field is specified as:[5]

$$q_v(z, t=0) = \begin{cases} q_v^{max} = 1.05 \times q_{v,s}, & d - L \times 0.5 \times 0.5 \le z < d + L \times 0.5 \times 0.5 \\ q_{v,e} = q_{v,s}, & elsewhere \end{cases}. \quad (19)$$

Above, $q_{v,e} = q_{v,s} = 3.872\ g/kg$ represents the clear-air region and $q_v^{max} = 4.066\ g/kg$ indicates the cloudy region. The length, $d \equiv L/2$ is the width of the cloud slab. The temperature field is initialized as:[5]

$$T(t=0) = \langle T(t=0)\rangle - 0.608\langle T(t=0)\rangle(q_v(z,t=0) - \langle q_v(t=0)\rangle)\,. \tag{20}$$

Equation (20) uses the information of initial water vapor field and assigns one temperature value to the cloudy region and another to the clear-air region. Based on Eqs. (19) and (20), the mean temperature and mean water vapor mixing ratio are not zero unlike the mean velocity. A Gaussian random field is superimposed on the initial temperature and water vapor fields, using the spectrum:

$$E_\theta(k) = \frac{16}{\sqrt{\pi/2}}\frac{\theta_0^2 k^4}{k_0^5}\exp\left(-\frac{2k^2}{k_0^2}\right), \tag{21}$$

where $\theta_0 = \langle T(t=0)\rangle \times 10^{-4}$ for the temperature field and $\theta_0 = \langle q_v(t=0)\rangle \times 10^{-3}$ for the water vapor field,[18] and $k_0 = 4.12k_{min}$. We simulate five cases for Scenario I, as summarized in Table II. These cases are generated by varying the level of momentum and thermodynamic forcing. The wavenumber bands corresponding to 'low' and 'high' levels of thermodynamic forcing are: $1 \le k_f/k_{min} \le 4.12$ and $1 \le k_f/k_{min} \le 16.18$, respectively. At the start of simulation, 16×10^6 dry aerosols are randomly placed in the cloudy region which gives a number concentration of 119 cm^{-3}. The size (radius) distribution of dry aerosols is assumed to be monodisperse with a mean of 0.1 μm for the five cases in Table II. The assumed dry aerosol parameters (119 cm^{-3} and 0.1 μm) are typical for stratocumulus clouds.[32] Dry aerosols absorb water from the environment to become aqueous aerosol particles which grow further and activate into cloud droplets in Scenario I.

TABLE II. Test cases for Scenario I and Scenario II.

Case	ϵ, TKE dissipation rate (m^2/s^3)	$\boldsymbol{F}_C$, Level of momentum forcing (m/s^2)	F_θ, Level of thermodynamic forcing (K/s or (g/kg)/s)	Initial particle size in Scenario I (μm)	Initial particle size in Scenario II (μm)
L	0.00063	Low	N/A	0.1	15
L-L	0.00063	Low	Low	0.1	15
L-H	0.00063	Low	High	0.1	15
LA	0.00063	Low	N/A	N/A	N/A
H	0.00723	High	N/A	0.1	15
H-L	0.00723	High	Low	0.1	15

Case	Description
L	Level of momentum forcing is low (L). Thermodynamic forcing is absent or $F_T = F_{q_v} = 0$.
L-L	Level of momentum forcing is low (L). Level of thermodynamic forcing is low (L).
L-H	Level of momentum forcing is low (L). Level of thermodynamic forcing is high (H).
LA	Level of momentum forcing is low (L). Thermodynamic forcing is absent. Particles are absent (A).
H	Level of momentum forcing is high (H). Thermodynamic forcing is absent or $F_T = F_{q_v} = 0$.
H-L	Level of momentum forcing is high (H). Level of thermodynamic forcing is low (L).

In Scenario II, the cloudy region is supersaturated ($S_e = 1.02$) but the clear-air region is subsaturated ($S_e = -0.9225$) at the initial time. The two regions correspond to $q_v^{max} = 3.949\ g/kg$ and $q_{v,e} = 0.30\ g/kg$, respectively. A total of 16×10^6 cloud droplets, each of them having a radius r of 15 μm, are randomly placed in the cloudy region at the initial time. Dry aerosols, also known as solutes, are assumed to be present in the core of cloud droplets with $r_d = 0.1\ \mu m$. For a known r_d, the r_c can be calculated using Eq. (13). The r_d and r_c are fixed at each particle position while the wet radius r evolves with time. We simulate five cases (Table II, except Case LA) in Scenario II which correspond to evaporation and deactivation of cloud droplets.

B. Parameters, grid resolution, and computing resources

Table III lists the parameters used in our model including the domain-averaged temperature, pressure, and Stokes number (S_t) at initial time. The chemical composition of dry aerosol is ammonium sulfate with a hygroscopicity (κ) value of 0.61.[33] Particles are allowed to grow (Eq. (11)) after the TKE dissipation rate becomes statistically stationary which takes 2 s. The input dissipation rates in the temperature ($\epsilon_{T,in}$) and vapor mixing ratio ($\epsilon_{q_v,in}$) fields and the Stokes numbers at initial time are shown in Table III. Note that the mean temperature, pressure, and density do not vary with height in our microscale model. A detailed analysis in this regard can be found in Ref. 6.

TABLE III. Parameters

Symbol	Value	Unit	Symbol	Value	Unit
q_v^{max}	4.066 (Scenario I) 3.949 (Scenario II)	g/kg	$q_{v,e}$	3.872 (Scenario I) 0.30 (Scenario II)	g/kg
L_h	2.5×10^6	J/kg	c_p	1005.0	J/kg/K
ρ_a	1.066	kg/m^3	ν	1.5×10^{-5}	m^2/s
ρ_l	1000	kg/m^3	$\langle p(t=0) \rangle$	82840	N/m^2
$\langle T(t=0) \rangle$	270.75	K	R_v	461.5	J/kg/K
k_T	0.0238	W/m/K	σ_l	0.072	N/m

$\mu_T = \mu_v$	2.2×10^{-5}	m²/s	M_l	0.018	kg/mol
R	8.314	J/mol/K	$\langle S_t(t=0) \rangle$	9.6×10^{-7} (Scenario I, L)	-
κ	0.61	-	$\langle S_t(t=0) \rangle$	2.16×10^{-2} (Scenario II, L)	-
$\epsilon_{T,in}$	1×10^{-5} (Scenario I)	K²/s	$\epsilon_{q_v,in}$	1×10^{-4} (Scenario I)	(g/kg)²/s
$\epsilon_{T,in}$	1×10^{-3} (Scenario II)	K²/s	$\epsilon_{q_v,in}$	1×10^{-2} (Scenario II)	(g/kg)²/s

As a compromise between grid resolution and computational cost, we use 256^3 grid points for all simulations. The $k_{max}\eta$ is a dimensionless parameter that is used to determine the sufficiency of grid resolution in DNS which is set to be at least greater than 1.[34] The $k_{max}\eta$ for the 'low' and 'high' turbulence levels are 2.46 and 1.43, respectively. The details of the numerical solvers are available in Ref. 6. The Perlmutter supercomputer, managed by the National Energy Research Scientific Computing Center (NERSC), United States Department of Energy, USA, has been used for the simulations.

IV. RESULTS AND DISCUSSIONS

We discuss the impact of external forcing on the fluctuating and mean thermodynamic fields in Sec. IV A. The impacts of thermodynamic forcing on condensation, aerosol activation, evaporation, and cloud droplet deactivation are explained in Sec. IV B and Sec. IV C. The concluding remarks in Sec. V summarize our findings.

A. The impact of external forcing on the thermodynamic fields

The temporal evolutions of the variance of temperature (σ_T^2) and vapor mixing ratio ($\sigma_{q_v}^2$) are shown in Figs. 2(a)-(d), respectively. We see that the fluctuating thermodynamic fields attain statistical stationarity when thermodynamic forcing is applied (Cases L-L, L-H, and H-L), but they decay otherwise (Cases L and H). A higher level of momentum forcing enhances mixing and phase change, resulting in a faster decay of the variance for Case H compared to Case L (Figs. 2(a)-(d)). On the other hand, thermodynamic forcing acts as a source of fluctuations and opposes the decay (Cases L-L, L-H, and H-L). The magnitude of the temperature fluctuations is larger in Scenario II (Fig. 2(c)) compared to Scenario I (Fig. 2(a)), as prescribed by the initial conditions (Sec. III A). The same is true for water vapor fluctuations (Fig. 2(d) vs Fig. 2(b)).

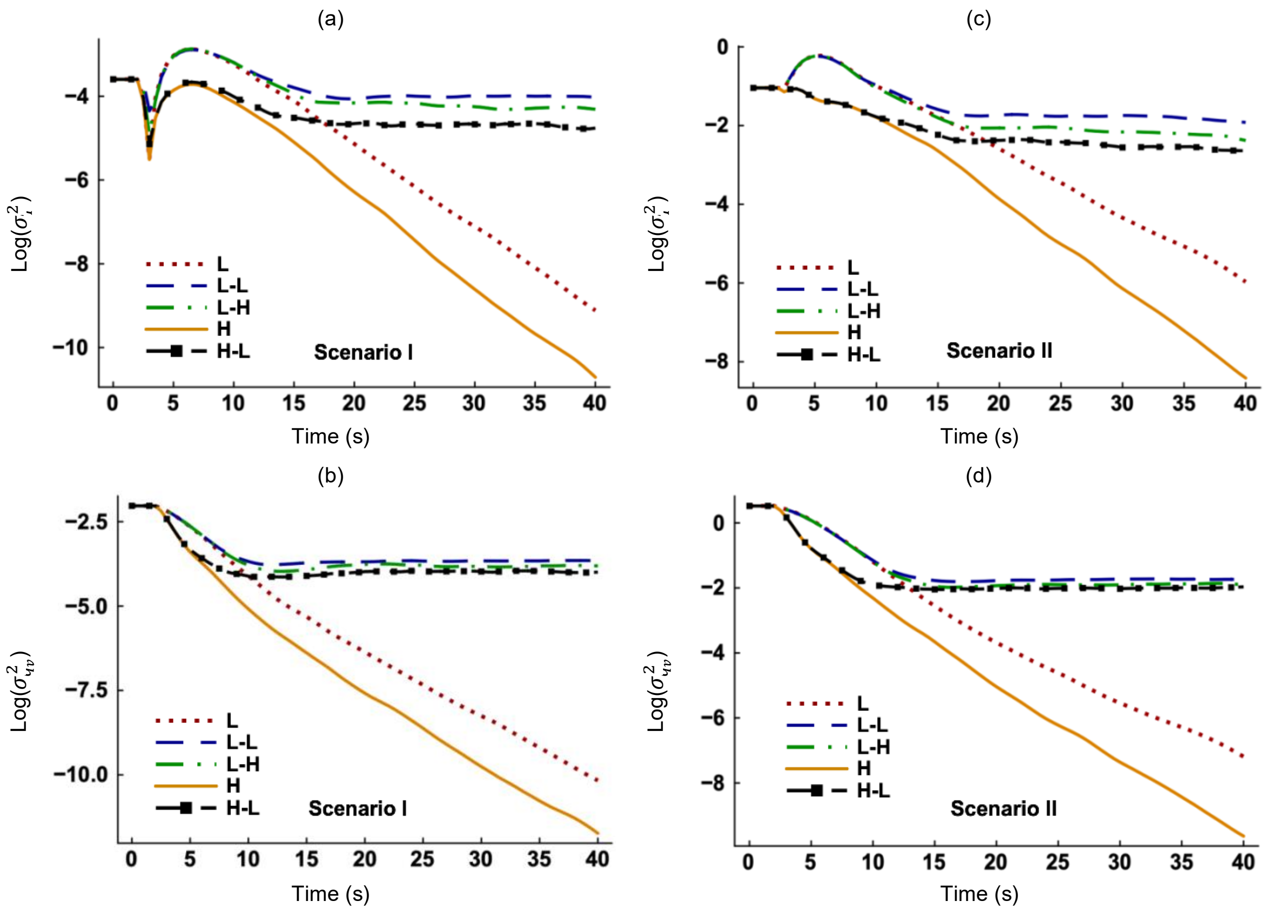


FIG. 2 Time evolution of the variances of (a) temperature and (b) water vapor mixing in Scenario I, respectively. Time evolution of the variances of (c) temperature and (d) water vapor mixing in Scenario II, respectively. The five cases shown are L, L-L, L-H, H, and H-L.

The energy spectra of temperature ($E_T(k)$) and water vapor mixing ratio ($E_{q_v}(k)$) are shown in Figs. 3(a)-(b) for Scenario I. The spectra of S_e resemble those of q_v and hence are not shown. For Cases L and H, where only momentum forcing is present, $E_T(k)$ and $E_{q_v}(k)$ differ at the large scales but converge at the small scales. This is expected because momentum forcing is applied at the large scales.[34] The integrals of $E_T(k)$ and $E_{q_v}(k)$ are 5.6×10^{-5} K^2 and 1.5×10^{-5} (g/kg)^2, respectively, for Case L. The corresponding values for Case H are 3.2×10^{-6} K^2 and 2.0×10^{-7} (g/kg)^2. This implies that thermodynamic fields contain more energy if the level of flow turbulence is low, consistent with the results in Fig. 2. There is a reduction in the downscale energy transfer at around $\log(k) \sim 3.0$ or $k \sim 1000$ before the final decay (Figs. 3(a)-(b)) when thermodynamic forcing is absent (L and H). With thermodynamic forcing (L-L, L-H, and H-L), more energy can be observed at all scales. The slopes (m) of the curves in the inertial subrange are shown in Figs. 3(a)-(b). We see significantly faster decay rate ($m = -4.25$) for Case L-H compared to Case H-L ($m = -2.9$) in Fig. 3(a). The decay rates for all cases fall between -4.25 and -2.6. These values suggest that the decay rate of the thermodynamic spectra is faster than that of Obukhov–Corrsin[35] ($-5/3$). Similar observation was made by Ladeinde and Oh[36] who studied the transport of a passive scalar in compressible homogeneous and isotropic turbulence. They reported a decay rate of -4 in the scalar field, which is within the range observed in the present study.

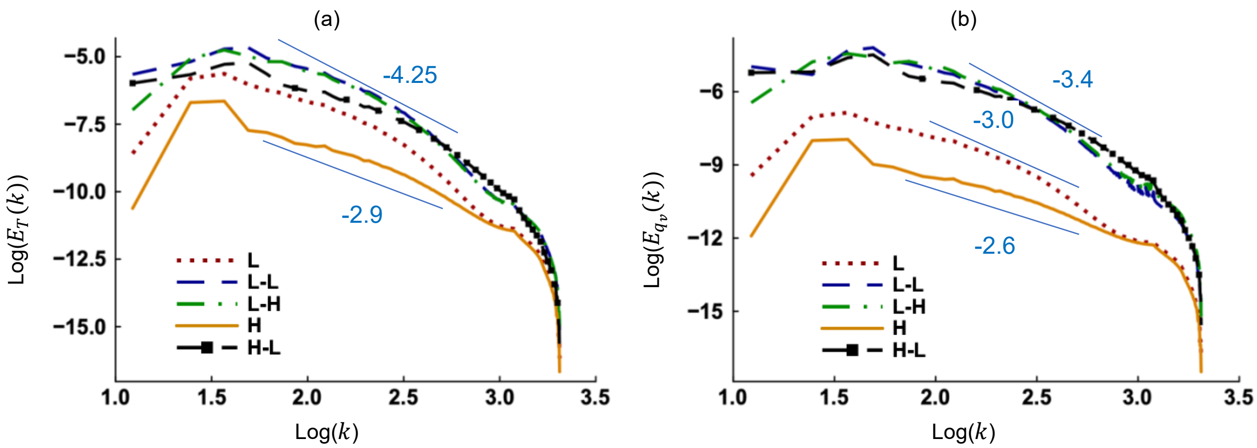


FIG. 3 Energy spectra of the (a) temperature and (b) water vapor mixing ratio for the five cases. Both the ordinate and abscissa are on a base-10 logarithmic scale. These results correspond to Scenario I at 20 s when thermodynamic fields are statistically stationary.

We further investigated the effects of phase change on $E_T(k)$ and $E_{q_v}(k)$, for which the two cases: L and LA are compared. Thermodynamic forcing is absent for both cases while additionally phase change is absent ($C_d = 0$) in the latter. We see in Figs. 4(a)-(b) that a reduction in downscale transfer occurs at $\log(k) \sim 3.0$ only if phase change is present (Case L). Otherwise, the spectra decay smoothly (Case LA). In addition, the presence of C_d changes the magnitudes of $E_T(k)$ and $E_{q_v}(k)$ at all scales. Since phase change affects the instantaneous particle radius (r), we show the energy spectra of r ($E_r(k)$) in Fig. 4(c). This quantity is relatively large at the small scales but negligible at the large scales. The backscatter[37] of the $E_r(k)$ clearly confirms the result that the phenomenon at small scale, i.e., condensation, is directly responsible for the growth of the large scale, or the increase in particle radius. While this makes a lot of sense in the physical space, the spectral analysis allows us to explain the phenomenon unambiguously.

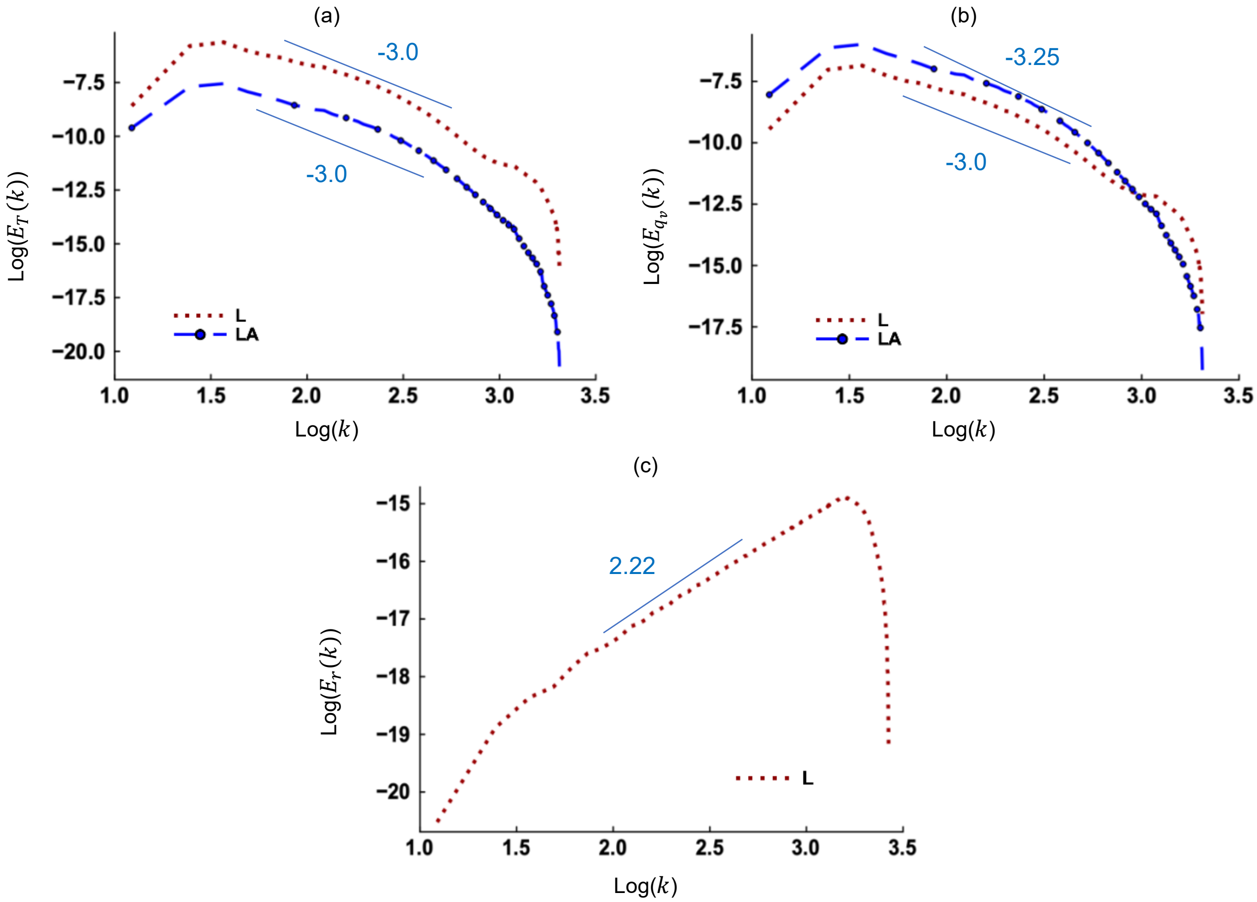


FIG. 4 Energy spectra of the (a) temperature and (b) vapor mixing ratio for Cases L and LA in Scenario I at 20 s. Energy spectrum of the (c) particle radii for Case L in Scenario I at 20 s. The slopes of the curves in the inertial subrange are also shown. Note that Case LA excludes phase change and has no $E_r(k)$.

The classical downscale energy transport is usually connected to the molecular diffusion at the small scales.[38] It seems then that the observed backscatter in Fig. 4(c) could be explained in terms of some negative molecular diffusion at the small scale. To substantiate this connection, we solve a steady-state transport equation for water vapor by introducing an effective diffusivity $\mu_{v,eff}$:

$$(\boldsymbol{u}.\nabla)q_v = -C_d + \mu_{v,eff}\nabla^2 q_v + F_{q_v} . \tag{22}$$

Figure 5(a) shows that $\mu_{v,eff}$ varies in space, with negative values at many points in the physical space. However, when viewed in the wavenumber space (Fig. 5(b)), the $E_{\mu_{v,eff}}(k)$ confirms the backscatter or the transfer of energy from small to large scales with a positive slope of 2.17. The suggestion in the present work that microscale phase change injects energy at the small scales (Fig. 4(c)) is supported by Jeffery[39] and Gotoh et al.[40] Their theoretical studies show that condensation or evaporation modifies the thermodynamic spectrum by creating a production region between the inertial and dissipation regions.

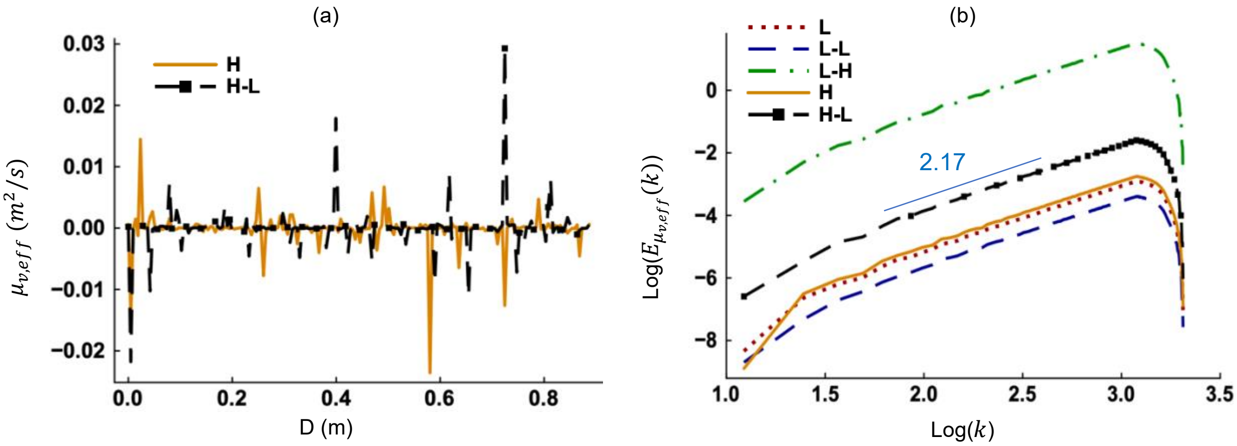


FIG. 5 (a) Spatial distribution of the effective vapor diffusivity for Cases L and L-H. b) Energy spectra of the effective vapor diffusivity for the five cases. These results correspond to Scenario I at 25 s. The results for only two cases are shown in Fig. 5(a) to keep the plot legible.

Fluctuations in the amount of water vapor (q_v') strongly influences the fluctuations in supersaturation (S_e') if thermodynamic forcing is applied. The spatial correlation coefficient between the two scalars can be defined as:[36]

$$R_{q_v' S_e'}(\boldsymbol{r}) = \frac{\langle q_v'(\boldsymbol{x},t) S_e'(\boldsymbol{x}+\boldsymbol{r},t) \rangle}{\sqrt{\langle \left(q_v'(\boldsymbol{x},t)\right)^2 \rangle \langle \left(S_e'(\boldsymbol{x},t)\right)^2 \rangle}} , \qquad \boldsymbol{r} = \Delta x \boldsymbol{i} + \Delta y \boldsymbol{j} + \Delta z \boldsymbol{k} , \tag{23}$$

where $\boldsymbol{r}$ is the vector of separation distance. We see in Fig. 6(a) that q_v' and S_e' are strongly correlated at short separation distances for the forced cases (L-L, L-H, and H-L), while the correlation is relatively weak for the unforced cases (L and H). However, the correlation decays faster for the forced cases which have a correlation length of approximately 0.1 m. As expected, for all cases, the correlation decays as the separation distance increases and significant memory no longer exists. The source term in Eq. (6) is the combination of external forcing and latent heat change, and its fluctuating component is $(F_{q_v} - C_d)' = F_{q_v} - C_d - \langle F_{q_v} - C_d \rangle$. Figure 6(b) shows that $R_{q_v'(F_{q_v}-C_d)'}$ is stronger for the three thermodynamically forced cases compared to two unforced cases. This observation is consistent with the finding (Fig. 2) that fluctuations are sustained when F_{q_v} is present. Figure 6(c) shows the cross-correlation between the fluctuations in supersaturation and vertical velocity (w') for Case L-L. We see in this figure that, S_e' and w' are not correlated, in agreement with the finding of Saito et al.[16] This observation also holds true for the other cases examined. The break of the link between S_e' and w' has previously been explained in terms of entrainment and turbulent mixing processes.[41]

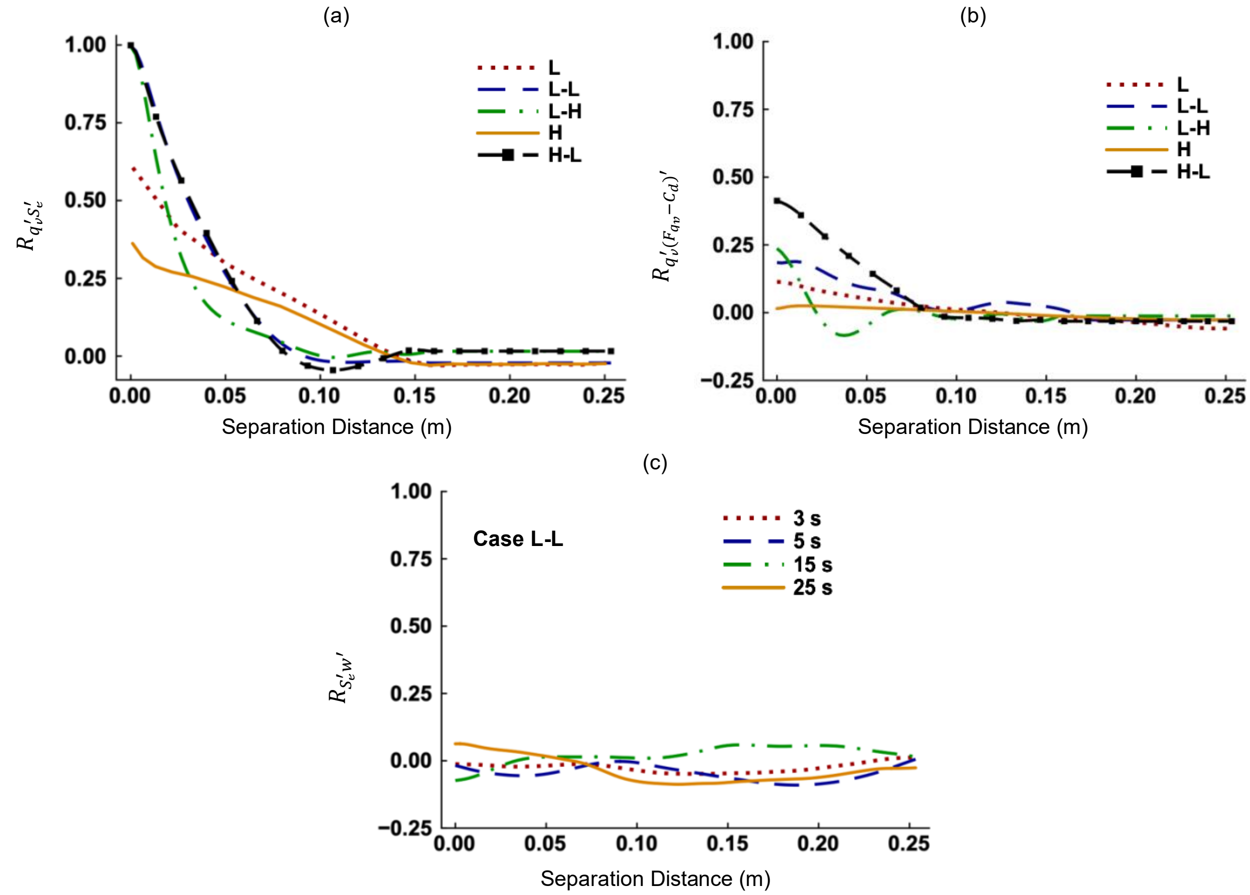


FIG. 6 (a) Correlation coefficient between fluctuations in the water vapor and supersaturation ($R_{q_v' S_e'}$). (b) Correlation coefficient between fluctuations in the q_v and $(F_{q_v} - C_d)$ which is denoted by $R_{q_v'(F_{q_v}-C_d)'}$. The plots represent the five cases in Scenario I at 25 s. (c) Correlation coefficient between fluctuations in the S_e and w ($R_{S_e' w'}$) for Case L-L in Scenario I at selected times.

We analyzed the probability distributions of the thermodynamic fluctuations when they are statistically stationary, using the data at 25 s in Scenario II. Figure 7(a) shows the PDFs of the fluctuating temperature normalized by the standard deviation (T'/σ_T), and Fig. 7(b) for the water vapor mixing ratio. The observations in Figs. 7(a) and 7(b) are representative of the other datasets. We see that the PDFs of T'/σ_T and q_v'/σ_{q_v} are not Gaussian. To measure the deviation from Gaussianity, we calculate the skewness (γ) and kurtosis (ω) of the distributions as[34]

$$\gamma = \frac{\langle(\varphi')^3\rangle}{\langle(\varphi')^2\rangle^{3/2}}\,, \quad \omega = \frac{\langle(\varphi')^4\rangle}{\langle(\varphi')^2\rangle^{4/2}}\,, \qquad \varphi = \frac{\theta'}{\sigma_\theta}. \tag{24}$$

where θ' is the fluctuating thermodynamic variable (T' or q_v') and σ_θ is the standard deviation. The values of γ and ω for the PDFs of temperature and water vapor are shown in Table IV. Note that skewness is 0 and kurtosis is 3 for a Gaussian distribution.

TABLE IV. Skewness and kurtosis values.

Case	Temperature skewness	Temperature kurtosis	Water vapor skewness	Water vapor kurtosis
L	0.25	2.94	0.22	2.87
L-L	−0.55	4.14	−0.86	4.56
L-H	−0.66	3.48	−0.82	4.05
H	0.22	2.09	0.21	2.09
H-L	−0.19	3.26	−0.62	3.80

We see in Table IV that both temperature and vapor PDFs are negatively skewed for the forced cases (L-L, L-H, and H-L), but are positively skewed for the unforced cases (L and H). Negative skewness implies that the distribution's tail extends toward smaller values and most values are larger than the mean. The kurtosis of temperature and vapor PDFs are greater than 3 for the forced cases (L-L, L-H, and H-L) but less than 3 for the unforced cases (L and H). Consequently, the PDFs have relatively sharp peaks when there is thermodynamic forcing (Figs. 7(a)-(b)) but have fairly rounded peaks[42] when there is no forcing. The non-Gaussian distributions of the thermodynamic fields have previously been reported for atmospheric clouds.[43,44] The largest value of T' is $\sim 4\sigma_T$ for Case L (Fig. 7(a)), $\sim 2\sigma_T$ for Case H, and $\sim 5\sigma_T$ for Cases L-L, L-H, and H-L. The largest value of q_v' is $\sim 3\sigma_{q_v}$ for Case L (Fig. 7(b)), $\sim 2\sigma_{q_v}$ for Case H, $\sim 5\sigma_{q_v}$ for Cases L-L and H-L, and $\sim 6\sigma_{q_v}$ for Case L-H. Anderson et al.,[45] in their experimental investigation of cloud processes, found that the largest value of T' is $\sim 4\sigma_T$ and that of q_v' is $\sim 6\sigma_{q_v}$ in the updraft region of a cloud. Agreement of the present simulations with the experimental results in Ref. 45 is evident.

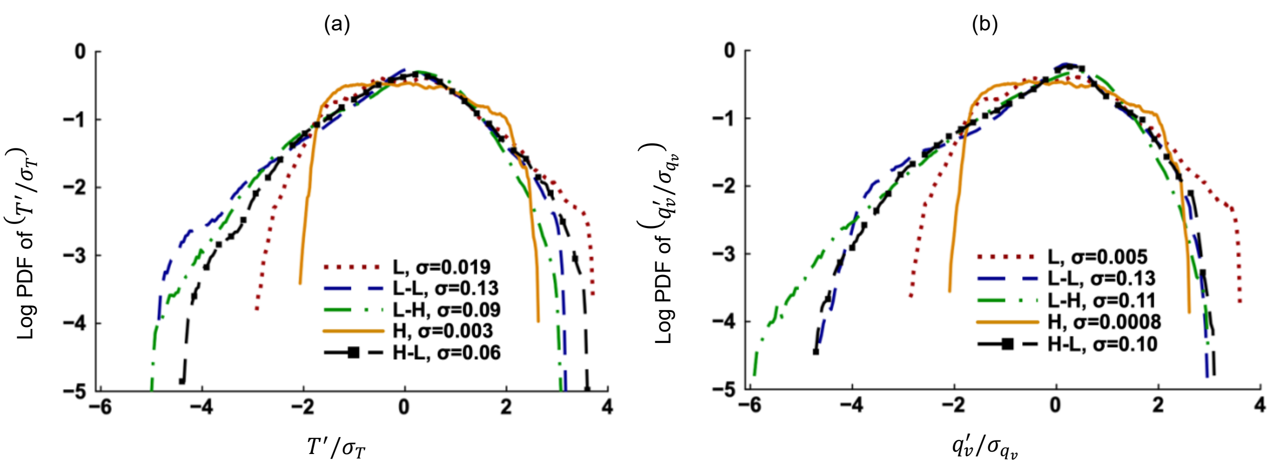
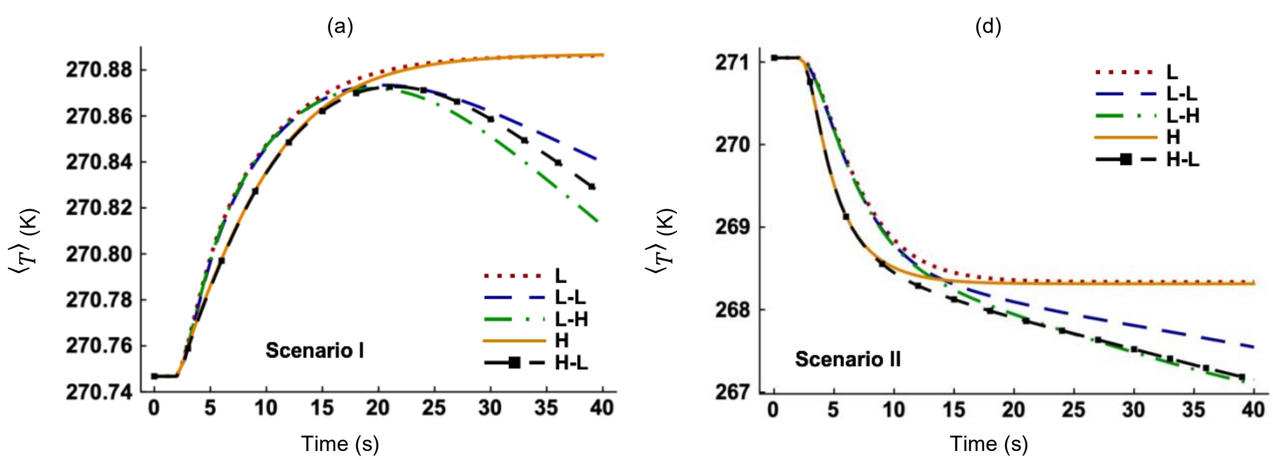


FIG. 7 Probability density functions (PDFs) of the (a) fluctuating temperature normalized by the standard deviation (T'/σ_T) and (b) fluctuating water vapor mixing ratio normalized by the standard deviation (q_v'/σ_{q_v}). The ordinate is on a base-10 logarithmic scale. The results correspond to Scenario II at 25 s. The PDFs of (S_e'/σ_{S_e}) resemble that of (q_v'/σ_{q_v}) and hence are not shown. The legends also list respective standard deviations.

Thermodynamic and momentum forcing influences the domain-averaged T and q_v. The mean temperature $\langle T \rangle$ increases during condensation (Fig. 8(a)) but decreases during evaporation (Fig. 8(d)) if thermodynamic forcing is absent (Cases L and H). In ambient clouds, these events correspond to condensational heating (C_d is positive) and evaporative cooling[46] (C_d is negative). The mean vapor mixing ratio $\langle q_v \rangle$ decreases in Scenario I (Fig. 8(b)) but increases in Scenario II (Fig. 8(e)) for Cases L and H, which is understandable because water vapor turns to liquid water during condensation and liquid water becomes water vapor during evaporation. The temporal evolution of the mean supersaturation $\langle S_e \rangle$ (Figs. 8(c) and 8(f)) resembles that of $\langle q_v \rangle$ (Figs. 8(b) and 8(e)) because $\langle S_e \rangle$ is linearly proportional to $\langle q_v \rangle$ (Eq. (14)). In the absence of thermodynamic forcing, the domain becomes saturated ($\langle S_e \rangle \sim 0$) in both scenarios, even though their initial conditions differ. The difference between thermodynamically forced and unforced cases becomes prominent after 20 s in Scenario I and 15 s in Scenario II. The changes in $\langle T \rangle$, $\langle q_v \rangle$, and $\langle S_e \rangle$ become asymptotic over time for the unforced cases. On the other hand, the temporal increase of $\langle T \rangle$ in Scenario I (Fig. 8(a)) is followed by a decrease if both the temperature and vapor fields are forced. The decrease conforms to the order: L-H > H-L > L-L, implying that a high level of thermodynamic forcing (Case L-H) enhances the decay of $\langle T \rangle$. The $\langle q_v \rangle$ decays similarly in Scenario I (Fig. 8(b)) and in Scenario II (Fig. 8(e)) for the three forced cases. Between Cases L and H, momentum forcing affects the $\langle T \rangle$, $\langle q_v \rangle$, and $\langle S_e \rangle$ until saturation ($\langle S_e \rangle \sim 0$) is reached but not afterwards. Stronger turbulence, which translates into larger values of eddy diffusivity, enhances mixing between the cloudy and dry air, leading to a higher evaporation rate and faster increase of $\langle q_v \rangle$ for Case H (Fig. 8(e)). However, strong mixing reduces cloud supersaturation, suppresses condensation, and slows down the decrease of $\langle q_v \rangle$ for the same case (Fig. 8(b)).



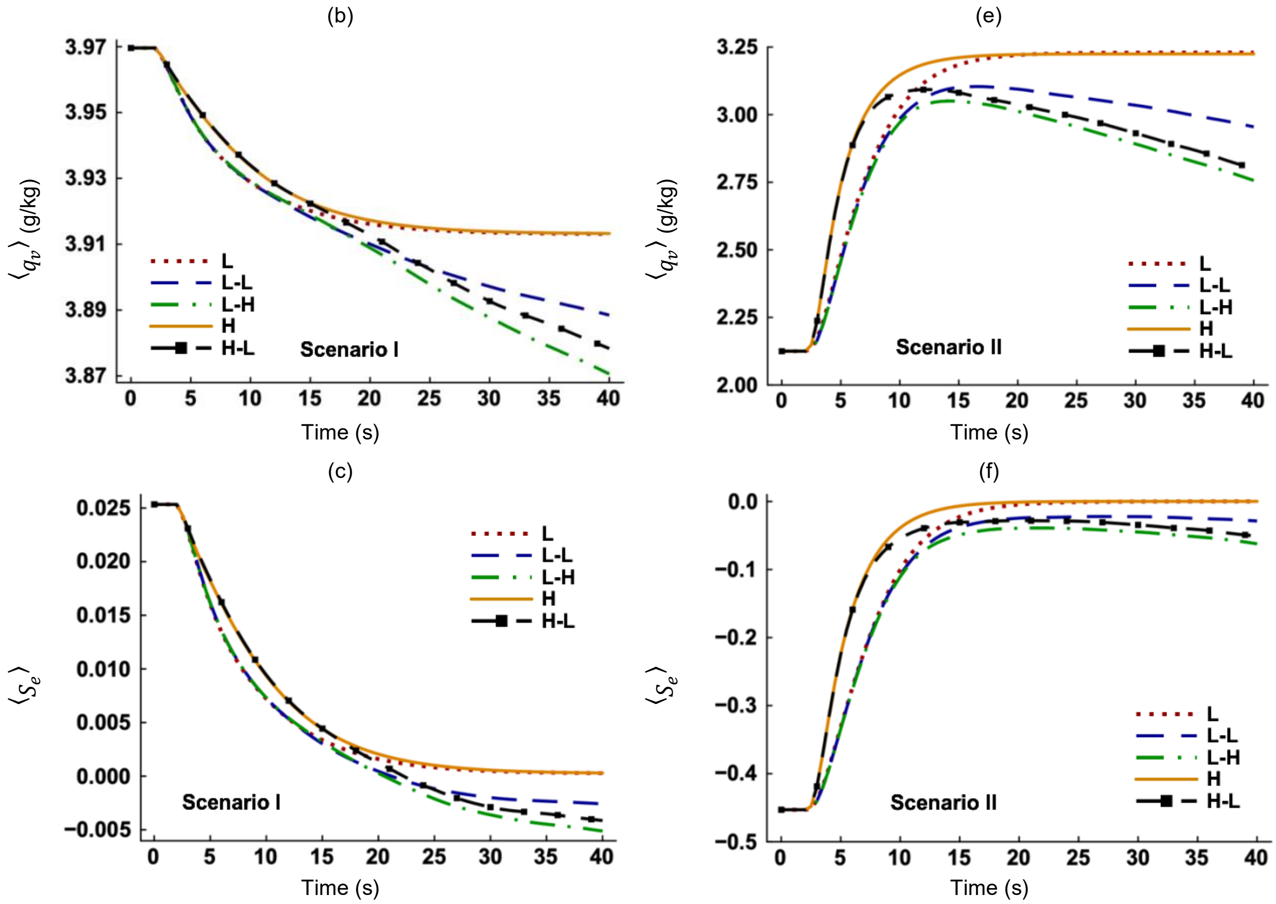


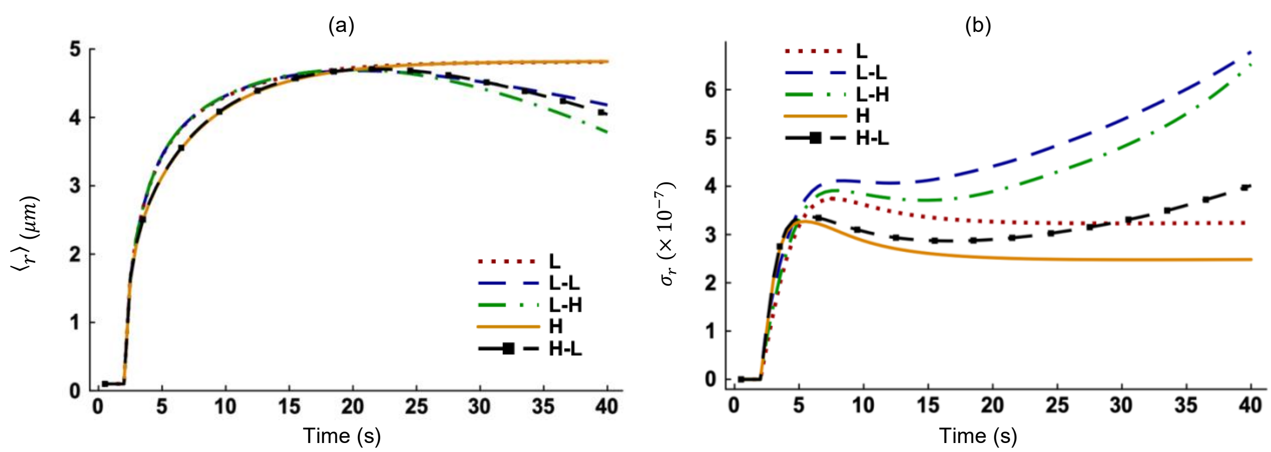
FIG. 8 Time evolution of the (a) mean temperature ($\langle T \rangle$), (b) mean vapor mixing ratio ($\langle q_v \rangle$), and (c) mean environmental supersaturation ($\langle S_e \rangle$), respectively in Scenario I. Time evolution of the (d) mean temperature ($\langle T \rangle$), (e) mean vapor mixing ratio ($\langle q_v \rangle$), and (f) mean environmental supersaturation ($\langle S_e \rangle$), respectively in Scenario II.

B. The impact of external forcing on cloud microphysics

1. Scenario I

The temporal changes in the mean radius ($\langle r \rangle$), its standard deviation (σ_r), and relative dispersion ($\sigma_r / \langle r \rangle$) of combined aerosol and cloud particles are shown in Figs. 9(a)-(c), respectively. The mean radius increases (Fig. 9(a)) exponentially at early times, corresponding to the positive supersaturation values in Fig. 8(c)). Particle growth levels off at approximately 25 s for the thermodynamically unforced cases (L and H) while the growth becomes negative for the forced cases (L-L, L-H, and H-L) after 22 s. Negative growth implies evaporative decay, which is slightly faster for a high level of thermodynamic forcing (L-H). The σ_r behaves differently and increases at early times for all five cases, decreasing with time for thermodynamically unforced cases (L and H) until becoming statistically stationary (Fig. 9(b)). When thermodynamic forcing is present, the increase in σ_r is followed by a slight decrease and then continuous increase. The relative dispersion of particle radii ($\sigma_r / \langle r \rangle$) indicates how broad or narrow the distribution of r is. We see that $\sigma_r/\langle r\rangle$ reaches a peak before gradually leveling off if thermodynamic forcing is absent (Fig. 9(c)). With thermodynamic forcing (L-L, L-H, and H-L), the temporal evolution of the $\sigma_r/\langle r\rangle$ resembles that of the standard deviation (σ_r), but with larger differences at later times due to the simultaneous increase of σ_r and decrease of $\langle r\rangle$. The relative dispersion is larger for the forced cases because sustained supersaturation fluctuations increase the σ_r. This also implies that the distribution of particle radii is broader with thermodynamic forcing. Many numerical models cannot produce particle size distributions as broad as that observed in experiments.[47] As noted by Liu et al.,[48] this discrepancy stems from the inadequate representation of turbulence-cloud microphysics interactions in those models. Our results clarify that the omission of thermodynamic forcing is a key inadequacy, resulting in a narrower particle size distribution.

Thermodynamic forcing also influences the activation and deactivation processes. As shown in Fig. 9(d), the total number of aerosol particles drops to near zero almost instantaneously for all cases after 2 s as all (aerosol) particles have grown beyond the critical radius of 1.26 μm and activated into cloud droplets. However, a close-up view reveals that some of the previously activated cloud droplets deactivate back into aerosols for two forced cases (L-L and L-H) at late times, unlike the cases without thermodynamic forcing. This corresponds to the evaporative decay reported in Fig. 9(a). These results suggest that although most cloud droplets remain activated, thermodynamic forcing could weaken activation.



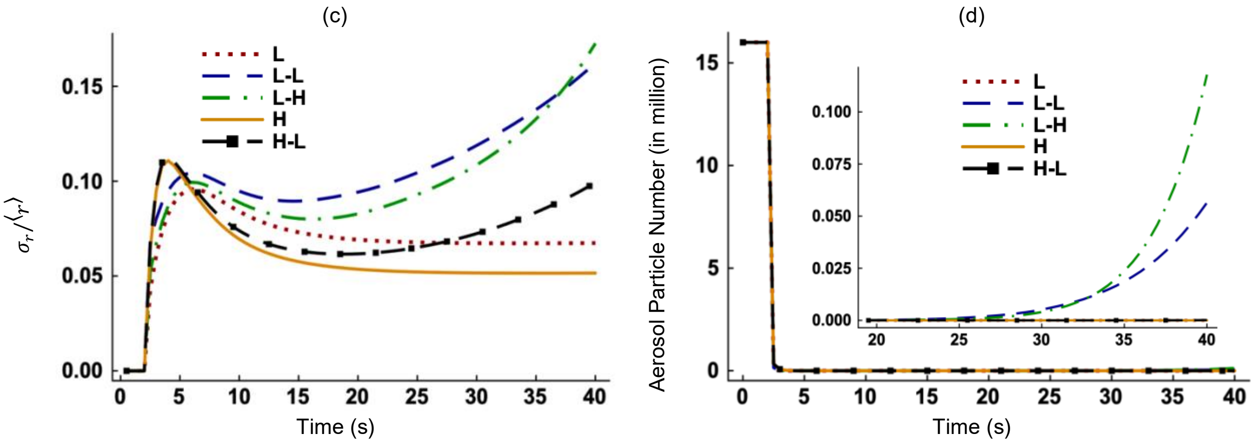


FIG. 9 Time evolution of the (a) mean radius ($\langle r\rangle$) where r includes both aerosol particles and cloud droplets, (b) standard deviation of particle radii (σ_r), (c) relative dispersion of particle radii ($\sigma_r/\langle r\rangle$), and (d) total number of aerosol particles in Scenario I. These results are produced using the Lagrangian data. The ordinate values do not change until particles are allowed to grow after 2 s.

To understand the complex effects of thermodynamic forcing on microphysics, we compare the particle size distributions (PSDs) for different forcing conditions at selected times (Figs. 10(a)-(d)). Note that PSD = PDF × n_c, where n_c is the particle number concentration. The PSD of monodisperse dry aerosol radii (0.1 μm) is represented by the dotted vertical line while that of the critical radii (1.26 μm) is indicated by the dashed vertical line. We see that all aerosol particles have activated into cloud droplets by 10 s for Case L (Fig. 10(a)). The corresponding PSD appears to be broader and negatively skewed. By 25 s, the PSD shifts to the right as cloud droplets grow bigger and the distribution narrows. No significant growth takes place between 25 s and 35 s, and the PSDs are identical (Fig. 10(a)). Chen et al.[49] has referred to this behavior as convergence of PSDs. Note that these results are consistent with the evolution of microphysical properties shown in Fig. 9. With increase in flow turbulence (Case H, Fig. 10(d)), the PSD is narrower at 35 s compared to Case L even though the mean radii are virtually identical for both cases at this time (Fig. 9(a)). The narrowing of PSD suggests increased homogeneity driven by higher turbulence intensity. In contrast, we see in Figs. 10(b)-(c) that thermodynamic forcing (Cases L-H and L-L) has a strong broadening effect on the PSD, especially at later times. A comparison of Case L (Fig. 10(a)) and Cases L-H and L-L (Figs. 10(b)-(c)) highlights three differences between the thermodynamically unforced and forced cases. First, although the maximum droplet radius increases marginally, the minimum droplet radius decreases considerably at 25 s and 35 s in the presence of thermodynamic forcing. Second, all particles are larger than the critical radius (r_c) at 10 s but some of them have become smaller than the critical radius at later times for Cases L-L and L-H. This suggests a change from activation to deactivation with thermodynamic forcing. Finally, unlike the thermodynamically unforced cases, for which the size distributions become narrower and converge with time, the PSDs for the forced cases broaden with time after a brief narrowing without trend of convergence.

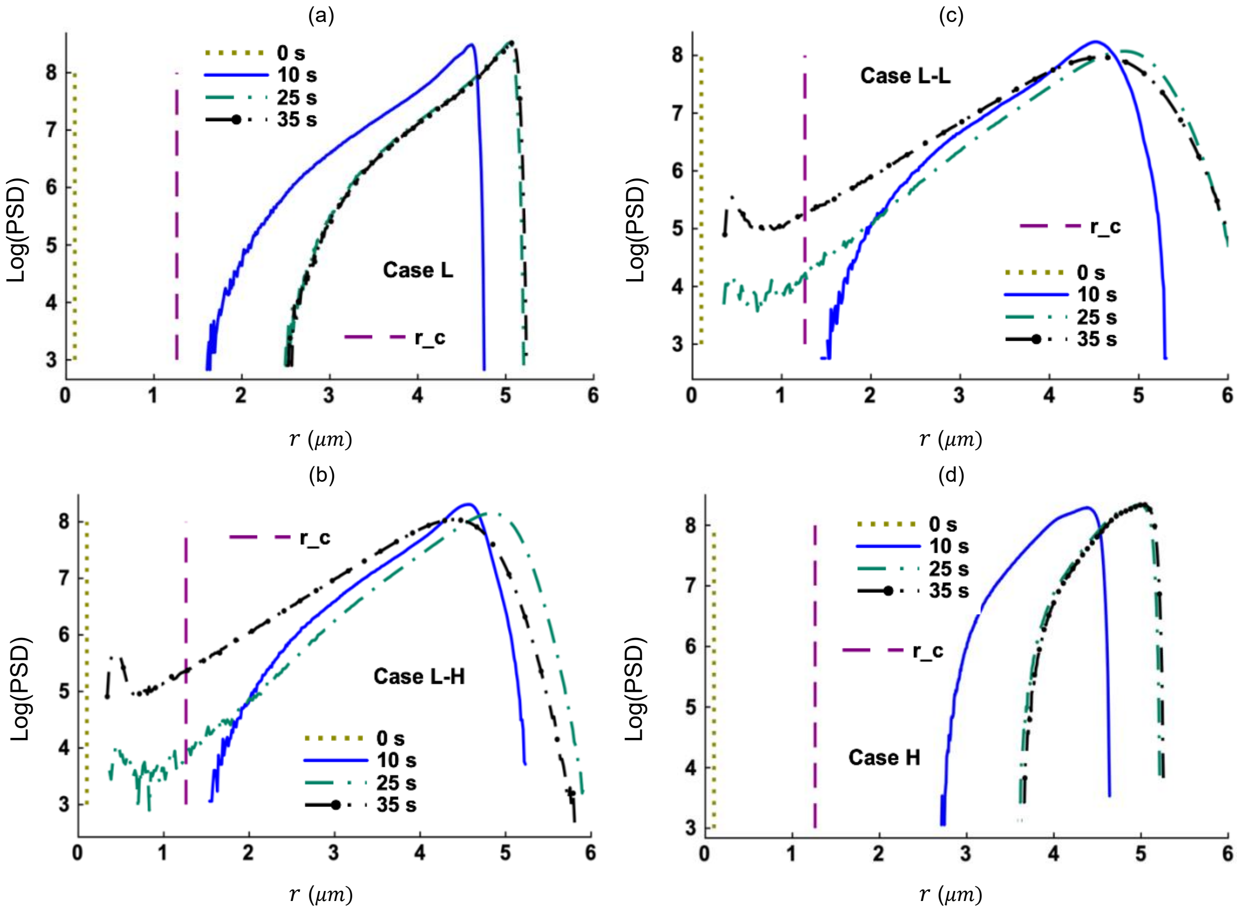
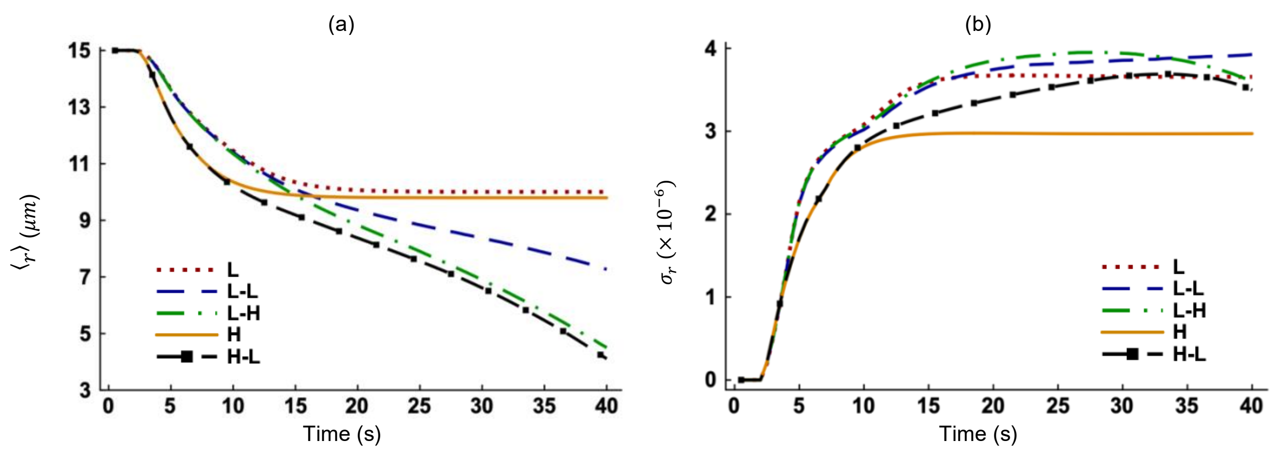


FIG. 10 Particle size distributions (PSDs) for (a) Case L, (b) Case L-H, (c) Case L-L, and (d) Case H at selected times in Scenario I. The ordinate is on a base-10 logarithmic scale. The critical radius (r_c) is shown in the dashed vertical line. The r in the abscissa includes both aerosol and cloud particles. The unit of PSD is $\mu m^{-1} cm^{-3}$.

2. *Scenario II*

The modeled cloud droplets for Scenario II initially have a monodisperse size distribution and a radius of 15 μm and they experience evaporative decay with time. Figure 11(a) shows that the mean radius decreases rapidly for the three thermodynamically forced cases (L-L, L-H, and H-L) compared to the unforced cases (L and H). This suggests that thermodynamic forcing enhances evaporation, and the larger the amount of forcing, the more the extent of evaporation (Case L-H compared to Case L-L, Fig. 11(a)). Evaporation is also enhanced by stronger turbulence (Cases H vs L). Evaporation is strongest when thermodynamic forcing is combined with a high level of flow turbulence (Case H-L). The standard deviation increases at early times for all cases. It levels off to an asymptotic value sooner (t~12 s) in the unforced cases (Fig. 11(b)). The $\sigma_r/\langle r \rangle$ keeps increasing for the forced cases (Fig. 11(c)) but levels off otherwise (Cases L and H). Chen et al.,[49] who used a particle-based DNS model to study entrainment-mixing without thermodynamic forcing, showed that $\sigma_r/\langle r\rangle$ becomes constant at long time. Increased momentum forcing enhances mixing and leads to more uniform particle growth. Consequently, the mean, standard deviation, and relative dispersion of particle radius are larger for Case L compared to Case H. However, for thermodynamically forced cases (L-L, L-H, and H-L), the changes in $\langle r\rangle$, σ_r, and $\sigma_r/\langle r\rangle$ do not follow any particular pattern. Several factors are responsible for this lack of consensus. First, there is a nonlinear interplay between momentum and thermodynamic forcing. The former promotes turbulent homogenization which is opposed by the latter (Fig. 2). Second, thermodynamic forcing sustains a non-zero, dynamically evolving supersaturation field (Fig. 8), which introduces variation in particle growth rate (Eq. (11)).

With evaporation and associated deactivation, the number of cloud droplets diminishes over time (Fig. 11(d)). The reduction is larger for Case L compared to Case H, implying that a higher level of flow turbulence suppresses deactivation in our model. With thermodynamic forcing, a higher level of turbulence corresponds to fewer deactivated cloud droplets (Cases L-L vs H-L). The order of deactivation (L-H > H-L > L-L > L > H) in Fig. 11(d) is not the same as the order of evaporation (H-L > L-H > L-L > H > L) in Fig. 11(a). Local subsaturation causes more droplets to fall below the critical barrier and deactivate, even if the bulk evaporation is low. Conversely, a higher rate of evaporation is possible even when many droplets do not deactivate.



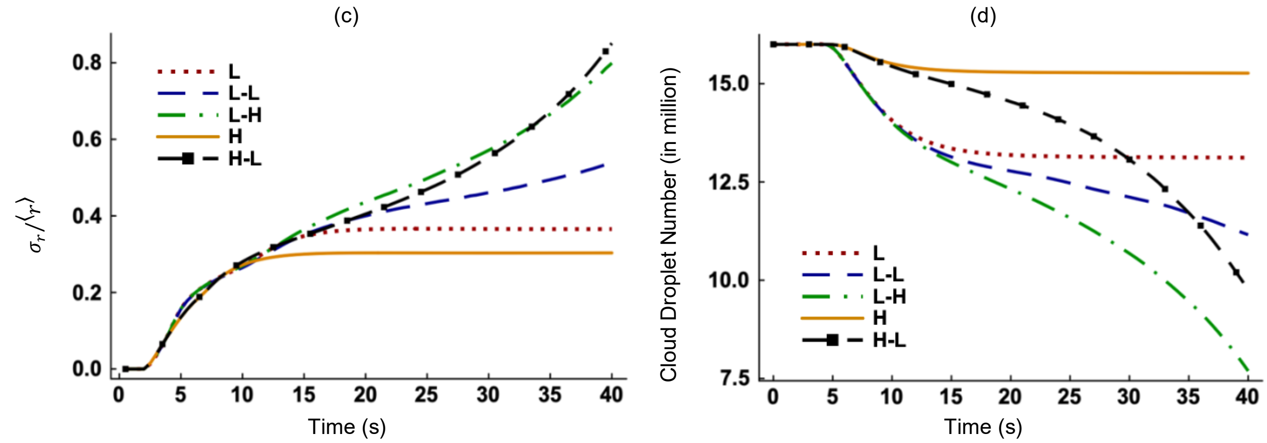
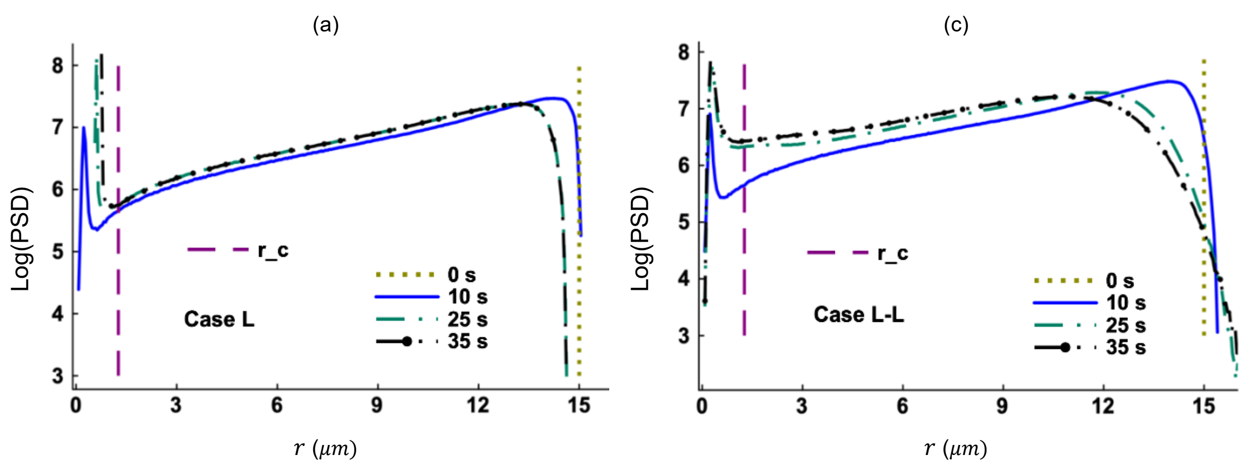


FIG. 11 Time evolution of the (a) mean radius ($\langle r\rangle$) of all particles (combined aerosol and cloud droplet), (b) standard deviation of particle radii (σ_r), (c) relative dispersion of particle radii ($\sigma_r/\langle r\rangle$), and (d) total number of cloud droplets in Scenario II. These results are produced using the Lagrangian data. The ordinate values do not change until particles are allowed to grow after 2 s.

The PSDs observed for Scenario II appear to be bimodal for all cases (Figs. 12(a)-(d)), with one peak to the left of the critical radius line and another to the right. Bimodality indicates inhomogeneous evaporation.[50] A group of cloud droplets experiences strong evaporation and deactivates into aerosol particles. This group appears on the left peak in Fig. 12, while the right peak consists of the rest of the droplet population. Manton[41] suggests that flow turbulence causes cloud droplets to grow (or shrink) at different rates, as the case may be, leading to a bimodal size distribution as seen here. For Case L, some droplets evaporate very little by 35 s and their sizes are very close to the initial size of 15 μm (Fig. 12(a)). If thermodynamic forcing is applied (Case L-L), a small number of cloud droplets grows instead of shrinking, thereby broadening the PSD (Fig. 12(c)) compared to the unforced case (L). With increased turbulence intensity (Case H), the PSD becomes narrower (Fig. 12(b)). More cloud droplets undergo complete evaporation if thermodynamic forcing is combined with a high level of turbulence (Case H-L), and the maximum droplet radius decreases at all the three time levels shown in Fig. 12(d).



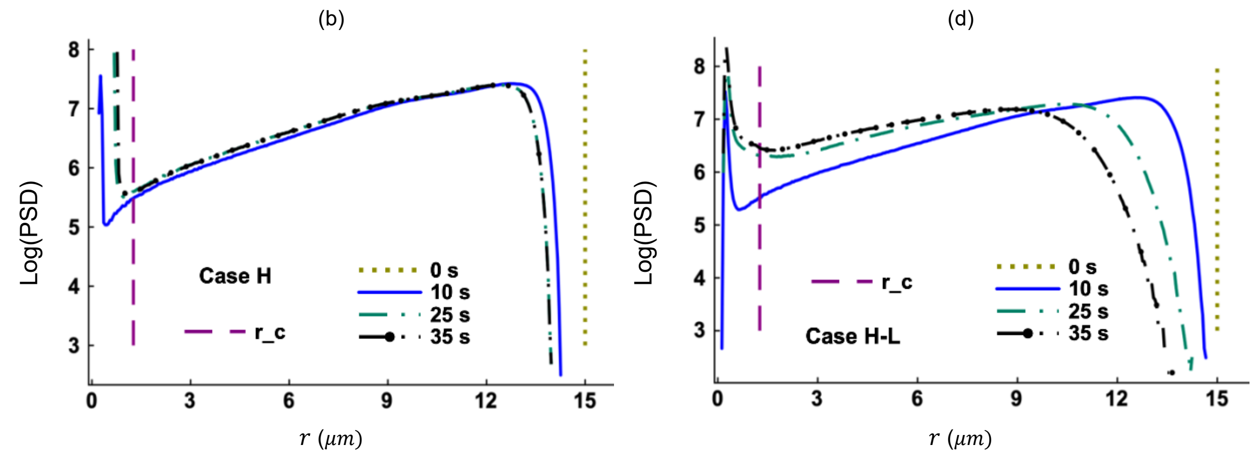


FIG. 12 Particle size distributions (PSD) for (a) Case L, (b) Case H, (c) Case L-L, and (d) Case H-L at selected times in Scenario II. The ordinate is on a base-10 logarithmic scale. The critical radius (r_c) is shown in the dashed vertical line. The r in the abscissa includes both aerosol and cloud particles. The unit of PSD is $\mu m^{-1} cm^{-3}$.

C. Further discussion on activation and deactivation

Prabhakaran et al.[51] identified three regimes of activation (and deactivation): i) mean-dominated, ii) fluctuation-mediated, and iii) fluctuation-dominated. These regimes were identified based on the comparison of the distributions of the critical supersaturation S_c and the environmental supersaturation S_e. In our previous study which excluded thermodynamic forcing,[6] only the mean-dominated regime was observed. However, as alluded to earlier, the inclusion of thermodynamic forcing caused activation and deactivation to occur in all three regimes. Additional details on this matter are provided below.

We select Case L-H in Scenario I for illustration and discuss the results. Figures 13(a)-(c) show the PDFs of environmental supersaturation S_e. The dotted and solid vertical lines represent the critical supersaturation S_c and the spatially-averaged environmental supersaturation $\langle S_e \rangle$, respectively. At 5 s, mean-dominated activation (MDA) takes place since $\langle S_e \rangle$ is much larger than S_c. At 15 s, fluctuation-dominated deactivation (FDD) and fluctuation-mediated activation (FMA) take place. Although $\langle S_e \rangle$ is slightly larger than S_c, fluctuations cause some S_e values to be smaller than S_c, leading to FDD. The FMA occurs when S_e is larger than S_c due to fluctuations. We see that $\langle S_e \rangle$ is slightly smaller than S_c at 25 s and activation is possible only through fluctuations, which translates into fluctuation-dominated activation (FDA). Fluctuation-mediated deactivation (FMD) occurs when fluctuations reduce S_e below S_c.

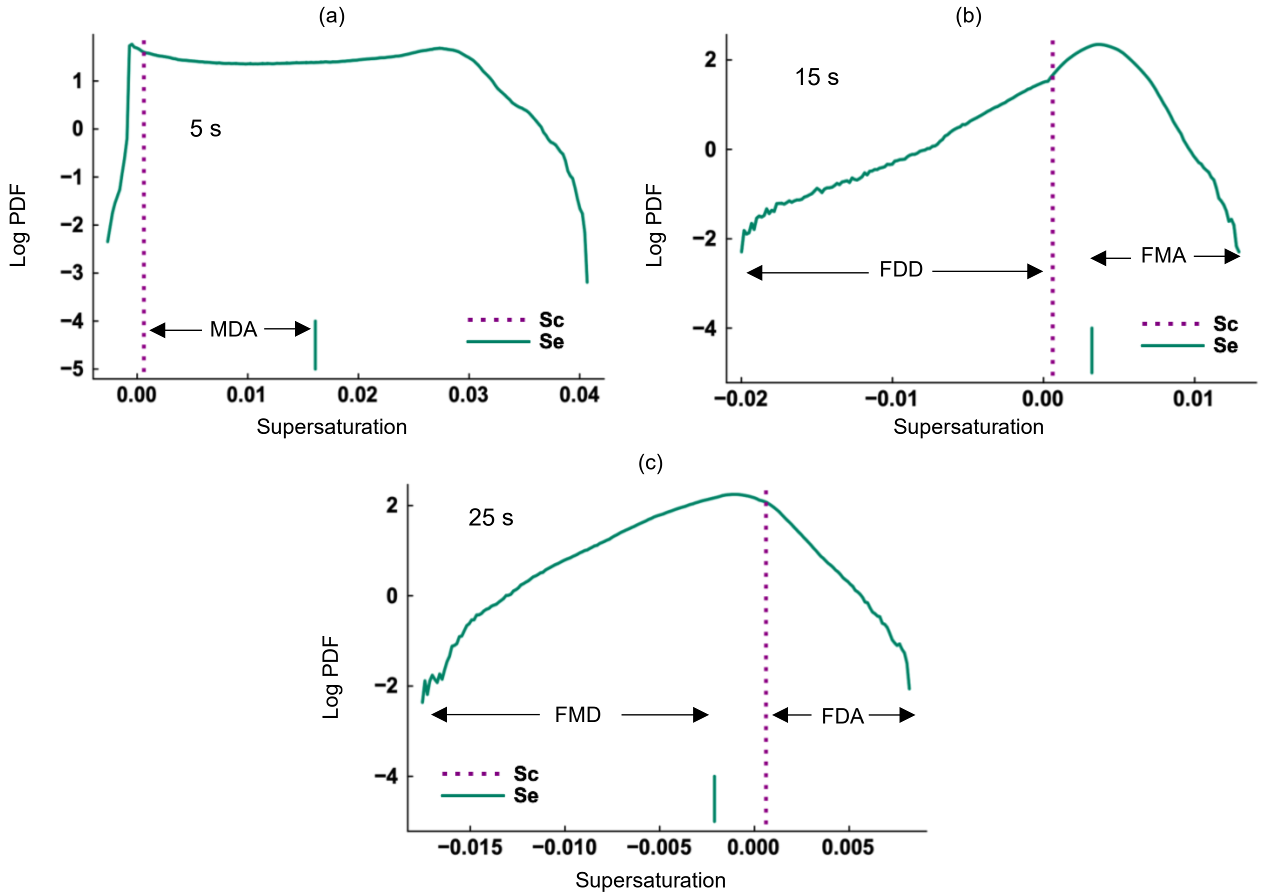
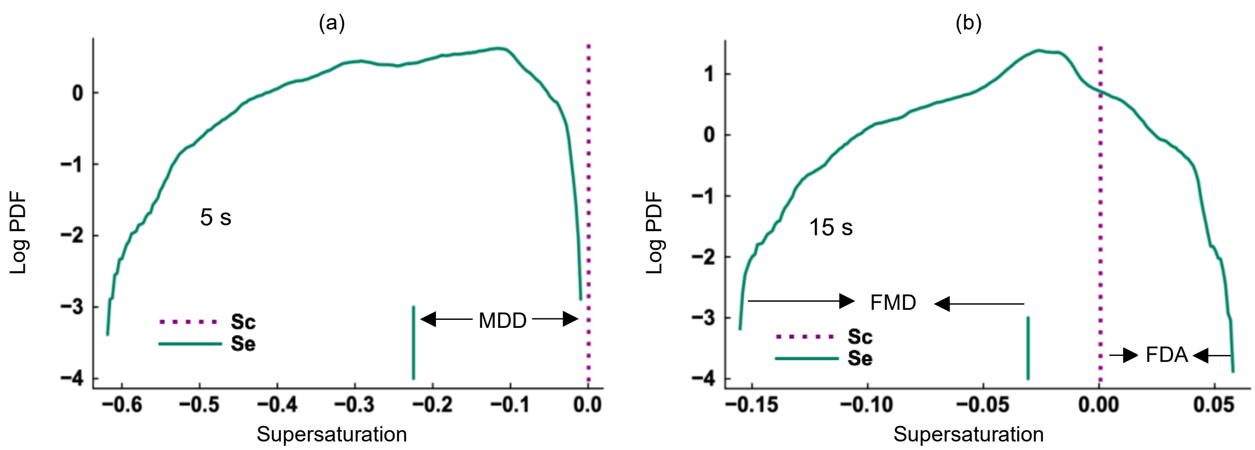


FIG. 13 Log PDFs of the environmental supersaturation (S_e) for Case L-H in Scenario I at (a) 5 s, (b) 15 s, and (c) 25 s. The dotted vertical line shows the value of critical equilibrium supersaturation (S_c). The solid vertical line indicates the mean (expected) S_e. MDA refers to mean-dominated activation, FDD is fluctuation-dominated deactivation, FMA is fluctuation-mediated activation, FMD is fluctuation-mediated deactivation, and FDA implies fluctuation-dominated activation.



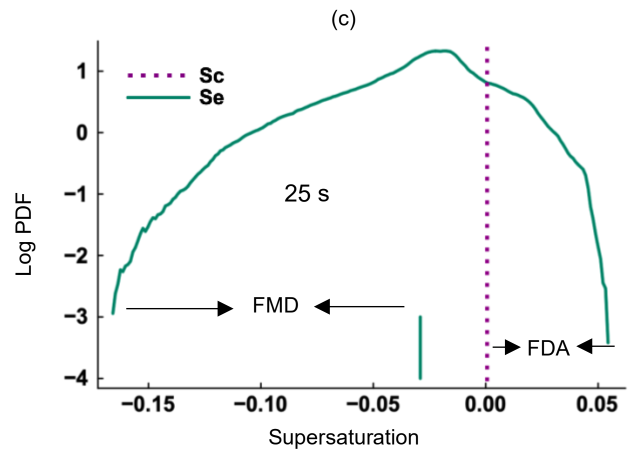


FIG. 14 Log PDFs of the environmental supersaturation (S_e) for Case H-L in Scenario II at (a) 5 s, (b) 15 s, and (c) 25 s. The dotted vertical line shows the value of critical equilibrium supersaturation (S_c). The solid vertical line indicates the mean (expected) S_e. MDD refers to mean-dominated deactivation, FMD is fluctuation-mediated deactivation, and FDA implies fluctuation-dominated activation.

Figures 14(a)-(c) show the PDFs of S_e for Case H-L in Scenario II. The S_c and $\langle S_e \rangle$ are also shown. At 5 s, mean-dominated deactivation (MDD) occurs because $\langle S_e \rangle$ is much smaller than S_c. At 15 s and 25 s, $\langle S_e \rangle$ remains smaller than S_c. But fluctuation causes some S_e values to be larger than S_c and support fluctuation-dominated activation (FDA). Fluctuation-mediated deactivation (FMD) is also identified in Figs. 14(b)-(c). The activation and deactivation regimes are summarized in Table V. The results in Figs. 13 and 14 emphasize that fluctuations in the S_e play an important role in determining the inter-conversion of aerosol particles and cloud droplets.

TABLE V. Different regimes of activation and deactivation.

Regime	Condition for activation	Regime	Condition for deactivation
MDA	$\langle S_e \rangle \gg S_c$	MDD	$\langle S_e \rangle \ll S_c$
FMA	$\langle S_e \rangle > S_c$ & $S_e > S_c$	FMD	$\langle S_e \rangle < S_c$ & $S_e < S_c$
FDA	$\langle S_e \rangle < S_c$ & $S_e > S_c$	FDD	$\langle S_e \rangle > S_c$ & $S_e < S_c$

V. CONCLUDING REMARKS

A three-dimensional particle-based DNS model is employed to study aerosol activation and droplet deactivation with low-wavenumber forcing in both the momentum and thermodynamic fields.[52] The influence of thermodynamic forcing on the thermodynamic and microphysical properties of clouds are examined. The key findings are as follows:

- The fluctuating thermodynamic fields become statistically stationary (Figs. 2(a)-(d)) if external thermodynamic forcing is applied but decay otherwise. The decay is caused by the lack of energy to maintain thermodynamic fluctuations. A relatively high level of momentum forcing accelerates the decay through stronger mixing and homogenization.

- Although thermodynamic forcing is applied at large scales only, it affects the energy spectra of thermodynamic fields at all scales (Figs. 3(a)-(b)) by interacting with the phase change term. The back transfer of energy from small to large scales is evident in the spectra of particle radius (Fig. 4(c)) and effective diffusivity (Fig. 5(b)).
- The backscatter of the $E_r(k)$ (Fig. 4(c)) confirms that microscale phase change, i.e., condensation, is directly responsible for the growth of the large scale, or the increase in particle radius.
- Velocity fluctuations do not have significant effects on supersaturation fluctuations (Fig. 6(c)). External thermodynamic forcing is the primary source of water vapor fluctuations (Fig. 6(b)).
- The PDFs of fluctuating temperature and water vapor fields are negatively skewed for the thermodynamically forced cases but positively skewed for the unforced cases (Figs. 7(a)-(b)). The kurtosis of temperature and water vapor PDFs are greater than 3 for the forced cases but less than 3 for the unforced cases.
- Thermodynamic forcing reduces the mean temperature, mean vapor mixing ratio, and mean supersaturation during condensation (Figs. 8(a)-(c)) and evaporation (Figs. 8(d)-(f)). If thermodynamic forcing is absent, the domain becomes saturated ($\langle S_e \rangle \sim 0$) and a thermodynamic equilibrium is maintained.
- In Scenario I, condensation and activation transition to evaporation and deactivation if thermodynamic forcing is present (Figs. 9(a) and 9(d)). The relative dispersion of particle radii ($\sigma_r / \langle r \rangle$) reaches an asymptote without forcing but grows continuously with forcing (Fig. 9(c)).
- The particle size distributions become narrower during condensation if the turbulence intensity increases and no thermodynamic forcing is applied (Fig. 10(d)). With thermodynamic forcing, a small fraction of the previously activated cloud droplets deactivates back into aerosol particles, and the PSDs broaden (Figs. 10(b)-(c)).
- Thermodynamic forcing enhances both evaporation (Fig. 11(a)) and deactivation (Fig. 11(d)). A high level of momentum forcing enhances evaporation (Fig. 11(a)) but suppresses deactivation of individual cloud droplets (Fig. 11(d)).
- The particle size distributions are found to be bimodal during evaporation (Figs. 12(a)-(d)). They are broader in response to thermodynamic forcing if the turbulence intensity is low (Fig. 12(c)). More cloud droplets evaporate completely if the turbulence level is high and thermodynamic forcing is present (Fig. 12(d)).

- Mean-dominated activation occurs at 5 s in Scenario I (Fig. 13(a)) and it transitions to fluctuation-dominated deactivation and fluctuation-mediated activation at 15 s (Fig. 13(b)). These regimes are followed by fluctuation-dominated activation at 25 s (Fig. 13(c)). Mean-dominated deactivation takes place at early times in Scenario II (Fig. 14(a)) and is followed by fluctuation-mediated deactivation and fluctuation-dominated activation at later times (Fig. 14(b)).

ACKNOWLEDGEMENTS

This study was jointly supported by the 2022 Stony Brook University (SBU)/Brookhaven National Laboratory (BNL) Seed Grant Program Award #: 94508, the BNL-Directed Research and Development Program (LDRD-25-008), and the U.S. Department of Energy's Atmospheric System Research (ASR) Program. The BNL is operated by the Brookhaven Science Associates, LLC, for the U.S. Department of Energy (DOE) under the Contract No. DESC0012704. This research used computing resources of the National Energy Research Scientific Computing Center (NERSC), a DOE Office of Science User Facility, provided by the award: NERSC ASCR-ERCAP0027399.

DATA AVAILABILITY

The data supporting the findings of this work are available from the corresponding author upon reasonable request.

APPENDIX A: THE IMPACT OF ADIABATIC LAPSE RATE

Previous DNS models such as those of Gao et al.[5] and Kumar et al.[3] ignored the adiabatic lapse rate term $(-g/c_p)w$ in the temperature transport equation because the rate of cooling with increasing altitude is insignificant if the domain size is small. To assess the impact of this adiabatic lapse rate term, we modify Eq. (5) as:

$$\frac{\partial T}{\partial t} + (\boldsymbol{u}.\nabla)T = \frac{L_h}{c_p} C_d + \mu_T \nabla^2 T - \frac{g}{c_p} w \,. \tag{A1}$$

Figures 15(a)-(d) show the temporal evolution of the $\sigma_{S_e}^2$, $\langle S_e \rangle$, $\langle r \rangle$, and $\sigma_r / \langle r \rangle$ for Cases L, L-L, and L-LR in Scenario I. Case L-LR solves Eq. (A1) instead of Eq. (5) and does not include water vapor forcing. We see that the decay of supersaturation variance is prevented for Case L-LR (Fig. 15(a)) but $\sigma_{S_e}^2$ is smaller than that for Case L-L by orders of magnitude. The profiles of $\langle S_e \rangle$, $\langle r \rangle$, and $\sigma_r / \langle r \rangle$ are identical for Cases L and L-LR (Figs. 15(b)-(d)). Thus, adiabatic lapse rate does not influence cloud microphysics in our DNS model. Therefore, Fourier forcing (Case L-L) is necessary to effectively maintain thermodynamic fluctuations and have desired variation in microphysical properties.

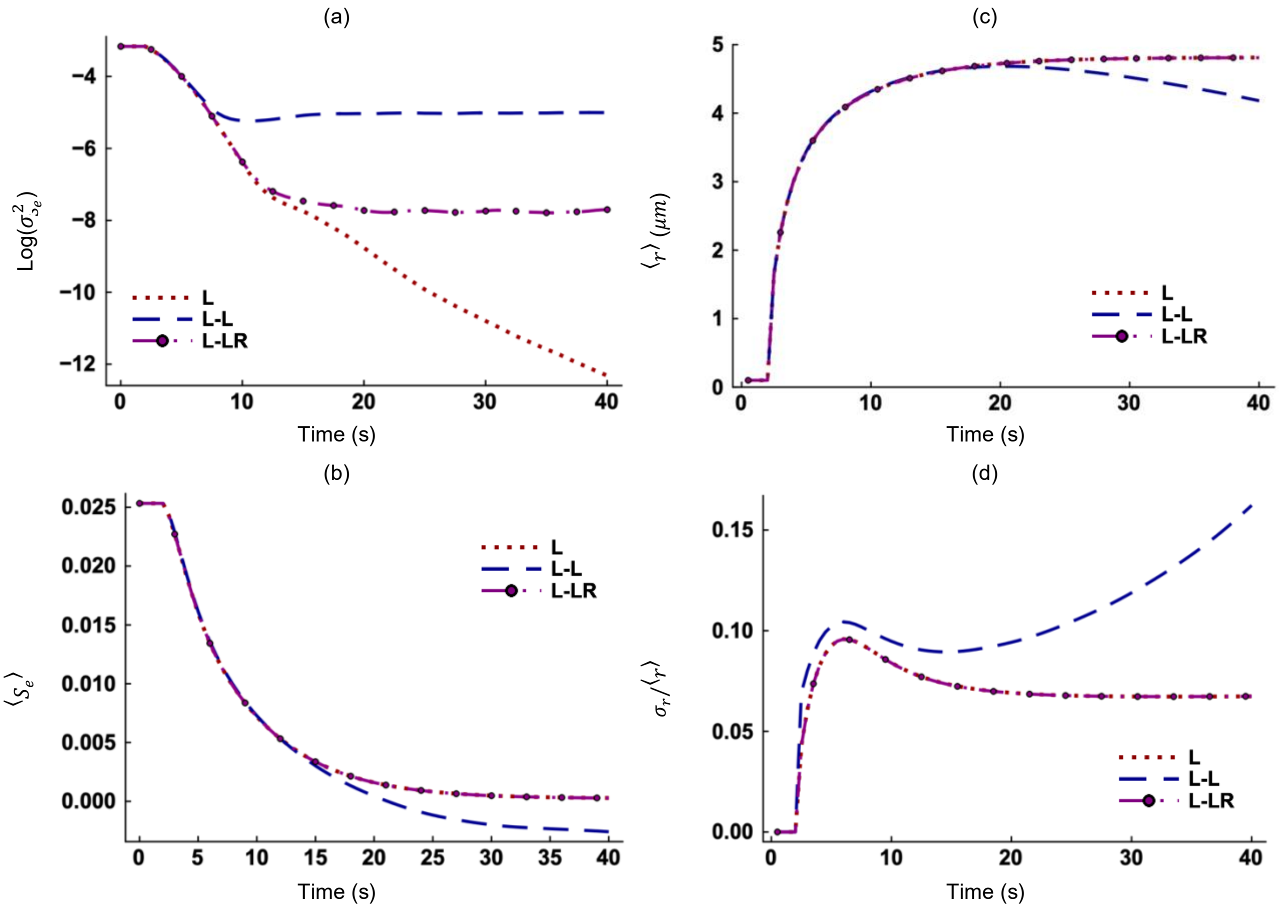


FIG. 15 Time evolution of the (a) variance of environmental supersaturation, (b) mean environmental supersaturation, (c) mean radius, and (d) relative dispersion of particle radii. The results represent Scenario I.

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